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Does Google Trends Improve Bitcoin Volatility Forecasts?

Evidence from sentiment-augmented GARCH and Random Forest models

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Abstract

Bitcoin is one of the most volatile financial assets in existence, making accurate volatility forecasts essential for risk management, derivative pricing, and regulatory oversight. This thesis provides a comprehensive comparison of classical econometric models and a machine-learning approach for forecasting Bitcoin volatility over the period January 2017 to September 2025. The analysis employs GARCH(1,1) and IGARCH(1,1) specifications under both normal and Student-t innovations, with and without augmentation by lagged Google Trends search interest as a proxy for investor attention, and these models are benchmarked against Random Forest regressions, both with and without sentiment features.

Using squared returns as the primary volatility proxy, out-of-sample performance is evaluated under the QLIKE loss function, and the Superior Predictive Ability (SPA) and Model Confidence Set (MCS) tests are applied to assess statistical significance. A series of robustness checks examine alternative volatility proxies (absolute returns), restricted evaluation windows, and additional sentiment lags.

The results show that the Gaussian GARCH(1,1) model consistently delivers the best forecast performance and that augmenting the variance equation with Google Trends sentiment produces statistically insignificant and economically negligible effects. The Random Forest models, even after modest hyperparameter tuning, underperform the benchmark GARCH specification, and the addition of sentiment features provides only marginal benefit. Both SPA and MCS tests confirm that no competing model significantly outperforms the GARCH-Normal benchmark.

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Chapter 1: Introduction

1.1 Background

Bitcoin has grown from being a mere technological experiment when it was first introduced in 2008/09 to becoming a leading globally traded financial asset with significant economic relevance. Since its introduction, Bitcoin has had phases of retail speculation, institutional experimentation, and just recently, there has been mainstream adoption of its trade even at the national levels as is the case of El Salvador where it was adopted by the state as an official currency(Argente & Van Patten, 2024). From 2017 through 2025, Bitcoin has been the market leader among cryptocurrencies and is constantly highly ranked in this class of assets as well as other top globally traded digital assets, with evolving regulatory developments as well as increasing liquidity which has resulted in enhanced adoption of Bitcoin as an asset. Bitcoin exhibits high volatility which makes it different from typical investment assets, with the variation in daily returns frequently exceeding those observed in equities, commodities, and foreign exchange markets, with huge price swings often influenced by market sentiment, investor attention, regulatory announcements, and the broader macro-financial developments. Given that this has become such an important asset, it raises the need to understand how to accurately forecast its volatility since Bitcoin holds highly significant value for both investors and regulators alike.

1.2 Importance of volatility forecasting

Volatility forecasting is an integral part of financial economics as the accurate predictions of conditional variance support in portfolio optimisation, hedging strategies, and risk management frameworks such as Value-at-Risk (VaR). For regulators and investors alike, volatility assessments enhance a deeper understanding of speculative bubbles, systemic risks, and market stability.

Conventional econometric models, especially the Autoregressive Conditional Heteroskedasticity (ARCH) model introduced by Engle (1982) and its generalised extension, GARCH, by Bollerslev (1986), remain the foundation of volatility modeling in financial economics since these models capture core empirical regularities such as volatility clustering and fat-tailed returns. These models, however, run into obstacles brought about by the fact that cryptocurrencies often exhibit stronger persistence, asymmetry patterns distinct from equities, and non-linear, sentiment-driven fluctuations compared to the highly seasonal data points like the national accounting statistics.

1.3 Research problem and research gap

While ARCH and GARCH models have been widely applied to cryptocurrencies for example, by Dyhrberg (2016); Chu et al. (2017); Katsiampa (2017), several limitations remain. First of all, many studies rely on short or outdated datasets ending before key structural events, such as the approval of U.S. spot Bitcoin ETFs in 2024(Gensler, 2024) which attracted more informed market participants to the Bitcoin spot market, possibly leading to a change in the composition of its investors, which resulted in a record net-inflows upwards of one and a half billion US dollars(Babalos et al., 2025). Secondly, few studies directly compare GARCH-family models with machine learning methods such as Random Forests, especially in the context of sentiment-driven volatility. This clearly outlines a gap given that investor behaviour has been demonstrated to influence greatly how Bitcoin performs as an asset(Katsiampa, 2017). It, therefore, necessitates the use of a more accurate variable representing sentiment, hence the role of sentiment, proxied by Google Trends and other search-based indicators, remains understudied over long time spans.

This raises the need for a systematic comparison of GARCH-family models and machine learning techniques, with and without sentiment variables, over an extended period of time.

1.4 Research objectives

The study has four main objectives. First, I compare the forecasting accuracy of GARCH-family models, particularly GARCH(1,1) and IGARCH(1,1) and Random Forests using Bitcoin daily data. Second, I evaluate the effects of incorporating sentiment, proxied by Google Trends, on volatility forecast accuracy, and next, I assess whether sentiment-augmented Random Forest models perform better than the traditional econometric approaches used to study volatility. Finally, I aim to provide an updated evidence on Bitcoin volatility dynamics using a comprehensive dataset encompassing major regime shifts.

1.5 The relevance of sentiment

Cryptocurrency markets, and in particular, crypto price performances, are heavily influenced by behavioural factors, media narratives, online discussions, and investor attention. Prior studies such as Mai et al. (2018) and Sapkota (2022) have shown that search intensity and social media conversations and trends can lead up to price volatility. In this sense, Google Trends which shows the popularity of search on Google by analysing the sample of real, anonymised searches, shows interest over time, by region, and in relation to other topics becomes a viable candidate to use as proxy for the investor attention, and is used in this study, and this integration of sentiment in the volatility models has the potential to capture otherwise unobservable market forces.

1.6 Structure of the thesis

The thesis is organised as follows. Chapter 1 introduces the background and motivation for studying Bitcoin volatility, outlines the research problem and objectives, and highlights the relevance of investor attention and sentiment in cryptocurrency markets.

Chapter 2 provides a comprehensive literature review. It begins with the foundational ARCH and GARCH literature, followed by the first-generation cryptocurrency volatility

studies, then examines asymmetry, smooth-transition dynamics, and realised volatility models. The chapter continues with behavioural and sentiment-driven research, including social media, news sentiment, search-based attention, and hybrid sentiment-econometric approaches, before concluding with the identification of the research gaps addressed in this study.

Chapter 3 describes the data sources, preprocessing steps, and the econometric and machine-learning methodology. It defines the return construction and volatility proxy, presents the GARCH(1,1), IGARCH(1,1), and their sentiment-augmented counterparts (GARCH-X and IGARCH-X), and details the Random Forest forecasting framework. The chapter also explains the maximum likelihood estimation procedure, the rolling-window out-of-sample forecasting design, and the forecast evaluation framework based on the QLIKE loss function, loss differentials, the Superior Predictive Ability (SPA) test, and the Model Confidence Set (MCS) procedure.

Chapter 4 presents the empirical findings. It starts with a detailed preliminary data analysis, including descriptive statistics, dependence and persistence diagnostics, distributional and stationarity tests, and correlation analysis. The chapter then reports the parameter estimates for the baseline GARCH and IGARCH models as well as their sentiment-augmented versions, followed by diagnostic tests (Ljung–Box and ARCH-LM). It concludes with the out-of-sample forecasting performance under the QLIKE loss function and includes the Random Forest results with in-sample fit and feature importance analysis.

Chapter 5 interprets the main empirical findings, discusses their consistency with the existing literature, and examines the limitations of the study. Particular attention is given to volatility persistence, the limited role of sentiment, the relative performance of econometric versus machine-learning models, and robustness across alternative volatility proxies and evaluation windows.

Finally, Chapter 6 summarises the key findings, outlines the empirical, methodological, and behavioural contributions of the study, discusses practical implications for investors and risk managers, and suggests directions for future research, including improved sentiment measures, hybrid models, high-frequency volatility modeling, and regime-based specifications.

Chapter 2: Literature review

2.1 Introduction

Over the past four decades, volatility forecasting has remained a central theme in financial econometrics, driven by the need to understand, quantify, and predict time-varying uncertainty in asset returns. The emergence of Bitcoin and other digital assets has renewed interest in this field, as cryptocurrency markets exhibit unique behavioural, structural, and technological characteristics that challenge traditional econometric approaches. Bitcoin, in particular, is characterised by extreme price swings, high-frequency reversals, heavy-tailed return distributions, and strong sensitivity to regulatory, technological, and behavioural influences. These features motivate comparative evaluations that combine classical econometric models with machine-learning techniques and sentiment-enhanced specifications.

This chapter provides a structured and comprehensive review of the academic literature on volatility forecasting, with a particular focus on Bitcoin and other cryptocurrencies. Section 2.2 revisits the foundational ARCH and GARCH literature, outlining their theoretical principles and empirical implications. Section 2.3 examines the first-generation cryptocurrency volatility studies that applied classical GARCH models to Bitcoin and other digital assets. Section **??** reviews contributions on asymmetry, smooth-transition dynamics, and realised volatility models. Section 2.5 discusses behavioural, sentiment, and attention-based research. Section 3.7 examines the growing literature on machine learning, with a particular focus on Random Forests and hybrid GARCH–ML approaches. The chapter concludes in Section 2.6 by identifying the main research gaps addressed in this thesis.

2.2 Foundational ARCH/GARCH canon

Financial return series often exhibit features inconsistent with classical homoskedastic linear models, and this necessitates the developments in the study of volatility and this inconsistency is often due to volatility clustering, heavy-tailed distributions, and serial dependence in higher-order moments. The foundational literature like the introduction of ARCH established the theoretical and empirical pillars that continue to support the contemporary financial econometric studies.

2.2.1 ARCH models

The seminal contribution of Engle (1982) introduced the Autoregressive Conditional Heteroskedasticity (ARCH) model which marked a major shift from constant-variance assumptions that had been prominent in prior literature. In Engle's specification, the conditional variance is modeled as a linear function of lagged squared residuals, a structure that captures the volatility clustering in which periods of high volatility followed by high volatility and produces fat-tailed unconditional return distributions even under normally distributed innovations. Engle's application to U.K. inflation demonstrated that macroeconomic uncertainty portrays strong temporal dependence and can be statistically modeled rather than treated as exogenous noise.

A number of key theoretical insights from the ARCH model came to the forefront. First, the comparison of conditional versus unconditional variance showed that volatility is forecastable from past shocks. In this regard, the ARCH processes were also shown to be leptokurtic, and with fat tails even when the shocks are Gaussian. And to check for the reliability of the ARCH model and whether it has unbiased results, Engle introduced the ARCH-LM test which still remains a standard diagnostic tool in econometrics.

In spite of the new pathways that the ARCH approach introduced for empirical finance, practical estimation still becomes cumbersome because capturing long memory requires many lags(ARCH(q) with large q), making estimation of these long memory models inefficient.

2.2.2 GARCH and IGARCH

To allow the conditional variance to depend on both the lagged squared residuals and lagged conditional variance terms, Bollerslev (1986) introduced the Generalized ARCH (GARCH) model, which addressed the practical limitations of high order ARCH. Thus, the GARCH(1,1) quickly became the primary model of volatility modeling due to its highly parsimonious and empirical effectiveness. The GARCH(1,1) model often captures long-range dependencies with only two parameters, α and β, addressing parsimony and to address volatility persistence, the model uses the sum of $\alpha+\beta$ to quantify the speed of mean reversion, and the model shows that unconditional distributions remain fat-tailed.

Subsequently, Engle & Bollerslev (1986) further extended the theory by introducing the Integrated GARCH (IGARCH) model, in which the persistence parameter $\alpha+\beta = 1$. In this specification, instead of volatility shocks decaying over time, they are allowed to accumulate indefinitely, implying that conditional variance behaves like a unit-root process. Ideally, IGARCH captures environments in which markets do not revert quickly to long-run volatility levels, and where new information, including small shocks, produces long-lasting effects on uncertainty.

The development of the IGARCH framework was particularly important for modeling exchange rates in the 1980s, where persistent volatility and slow mean reversion were empirically pervasive. In subsequent years, researchers found that cryptocurrencies exhibit remarkably similar characteristics: high persistence, weak mean reversion, and volatility that remains elevated long after major shocks. IGARCH, thus, serves as a bridge between classical GARCH methodologies and the empirical reality of digital asset markets, in which volatility dynamics often resemble near-unit-root behaviour.

2.2.3 Continuous-time links

To bridge the discrete-time GARCH with the continuous-time stochastic volatility models, Nelson (1990) introduced the exponential GARCH(EGARCH), showing that as the sampling intervals shrink, many GARCH processes converge to diffusion processes. This discovery provided the theoretical justification for using GARCH in contexts traditionally

dominated by continuous-time models, such as option pricing. Nelson equally proved that GARCH(1,1) converges to a process with an inverted-gamma stationary distribution, while exponential-GARCH processes yield log-normal variance diffusions, thus providing more insights into the theoretical legitimacy for applying GARCH to high-frequency data and enriched the versatility of volatility models.

2.2.4 Stylised facts

In the subsequent years, Bollerslev et al. (1994) defined a set of stylised facts about financial return series after surveying more than a decade of empirical research, showing that these returns remain stable across assets, markets and time periods. These findings still serve as a guide for the specification, estimation and evaluation of volatility models in the GARCH tradition and beyond.

Volatility clustering is an essential fact associated with financial asset returns as large price movements tend to be followed by large movements, either negative or positive, while periods of stability and peace tend to persist. This temporal dependence in the conditional variance tends to be one of the strongest characteristics of financial returns and forms the core empirical motivation for GARCH models as it suggests that volatility is persistent and forecastable, contradicting the notion of constant variance assumed in early asset pricing models prior to the introduction of the ARCH framework.

Bollerslev and the co-authors identified that leverage effects are integral in financial markets since in these markets, volatility responds asymmetrically to news and bad news, in particular, with negative returns typically increases volatility more than good news. The authors attribute this phenomenon to financial leverage, highlighting that a fall in equity value increases leverage and raises risk, or to behavioural factors such as asymmetric reactions to losses. On this basis, even though leverage effects are a defining feature in equities, they are notably weaker or even absent in Bitcoin and many cryptocurrencies, highlighting the fundamental differences in market microstructure.

Bollerslev and the co-authors found that empirical return distributions exhibit kurtosis far exceeding that of the normal distribution, meaning that the results are heavily fat-tailed and the financial returns display more frequent extreme movements than the Gaussian

models predict as the GARCH processes with conditional normal errors can already generate heavy-tailed unconditional distributions, a feature that is further amplified when using fat-tailed innovation distributions like the Student-t.

The financial markets are found to rely heavily on information flow, and they experience predictable spikes in volatility around scheduled events such as monetary policy announcements, labour market releases, or earnings reports. Whereas these predictable bursts do not imply that the effect of information is neither uniform nor random, they have motivated the introduction of models that explicitly incorporate exogenous information or calendar effects, features that are particularly relevant to cryptocurrencies, where regulatory announcements, exchange hacks, and protocol updates often generate disproportionate volatility.

Bollerslev et al. (1994) found the existence of volatility co-movements across assets, noting that the volatilities across different assets, sectors, or markets tend to move together, which shows the presence of common shocks or shared macroeconomic risks that necessitates the need for multivariate GARCH models such as CCC, DCC, and BEKK, and explains why systemic or contagion effects can emerge during periods of market stress. This co-movement is also observed across cryptocurrencies, which often respond simultaneously to global risk sentiment or major news events.

These stylised facts jointly form the conceptual backbone for modern volatility modeling, justifying the need for dynamic conditional variance models like GARCH, as well as more flexible nonlinear approaches such as regime-switching, smooth-transition models, squared returns frameworks, and machine learning methods. The persistence, asymmetry, tail behaviour, and cross-asset dependence described by these facts are very present in cryptocurrency markets, making them crucial considerations in any attempt to model Bitcoin volatility.

The stylised facts documented by Bollerslev et al. (1994) provided the theoretical and empirical foundation for modern volatility modeling across traditional asset classes. When Bitcoin emerged as a major financial asset after 2017, researchers naturally turned to this well-established GARCH toolkit to test whether the same regularities—volatility clustering, persistence, heavy tails, and asymmetry—held in the new cryptocurrency market. The first

generation of empirical studies (2016–2019) therefore applied classical ARCH, GARCH, and IGARCH specifications to Bitcoin and other digital assets, seeking to establish whether this novel asset class behaved like equities, commodities, or FX, or whether it required fundamentally new modeling approaches.

2.3 First generation cryptocurrency volatility studies

The initial research on cryptocurrencies applied the classical GARCH-family models with the aim of assessing whether Bitcoin behaved like traditional assets or not, identifying important empirical characteristics like extreme volatility, persistence patterns, and possible regime shifts which have served as defining features for the subsequent research in this area.

2.3.1 Symmetry and volatility persistence

To position the asset within the broader volatility landscape, Dyhrberg (2016) carried out one of the earliest thorough GARCH applications to Bitcoin, comparing its behaviour to gold and the U.S. dollar. Using both GARCH(1,1) and EGARCH specifications, the study highlighted several important features that have shaped much of the early cryptocurrency volatility literature.

First, Dyhrberg discovered the presence of high volatility persistence similar to commodity markets, implying that shocks to Bitcoin variance disappear slowly and can influence volatility over extended periods, a finding which has been found to be consistent with the later findings as in Chu et al. (2017) and Katsiampa (2017), which frequently identified IGARCH-type behaviour for Bitcoin and other cryptocurrencies.

Second, the existence of symmetric volatility responses, which indicate that positive and negative return shocks have similar effects on the volatility of Bitcoin, sharply contrasts with equity markets, where negative shocks typically trigger larger volatility increases. In their subsequent study, Cheikh et al. (2020) confirmed this phenomenon and even documented *inverted asymmetry* for certain cryptocurrencies, where volatility sometimes increases more after positive news, a feature contradicting the traditional leverage effect in financial markets.

Finally, since Bitcoin exhibited weak correlations with other equities, Dyhrberg (2016)

suggested that it can be used potentially as a diversifying asset and hedge against equities, thus assuming roles similar to the precious metals like gold, which sets the stage for exploring how Bitcoin as an asset behaves in relation to other assets as well as macroeconomic and behavioural variables.

Dyhrberg's results accordingly imply that the absence of leverage effects reflects Bitcoin's distinctive behavioural and informational environment, characterised by retail-driven sentiment, speculative trading, and limited balance sheet mechanisms compared to equities, a fact consistent with later studies showing that sentiment and attention variables such as forum discussions as presented by Mai et al. (2018), news narratives by Sapkota (2022) and Coulter (2022), and Google search intensity play a first-order role in driving volatility.

The results also show that, the symmetry of volatility responses suggested early on that classical GARCH extensions designed for leverage effects (e.g., EGARCH, GJR-GARCH) might be less relevant for Bitcoin. Indeed, empirical comparisons in Chu et al. (2017) and Katsiampa (2017) often found that symmetric models (GARCH, IGARCH, CGARCH) performed as well or even better than asymmetric ones.

Finally, Dyhrberg's work established Bitcoin as an asset whose volatility behaviour is closer to commodities than equities, thereby reinforcing the need for models capable of capturing persistent, sentiment-sensitive, and regime-dependent dynamics which serves as the basis for later research using regime-switching GARCH models (Ardia et al., 2019), smooth-transition models (Cheikh et al., 2020), and hybrid machine learning approaches (Zahid et al., 2022; Han et al., 2025), all of which attempt to account for nonlinearities and evolving market conditions not captured by early single-regime models.

2.3.2 Comprehensive GARCH comparisons

In carrying out one of the first large-scale, systematic comparisons of the GARCH-type models applied to digital assets, Chu et al. (2017) estimated twelve volatility specifications, including GARCH, EGARCH, GJR-GARCH, FIGARCH, and IGARCH across seven major cryptocurrencies, with results revealing several distinctive features of cryptocurrency volatility dynamics. In the first place, IGARCH frequently emerged as the best model for modeling the volatilities of cryptocurrencies, suggesting that volatility shocks in Bitcoin

and other cryptocurrencies exhibit near-unit-root persistence, a finding consistent with the long-memory properties of Bitcoin which was later explored through component and regime-switching models. Secondly, normal innovations were found to often outperform heavy-tailed alternatives, a somewhat surprising result given the strong empirical evidence of fat tails in traditional equity and commodity returns. This result may reflect the dominance of structural breaks, speculative bursts, and regime shifts in crypto markets rather than heavy-tailed shock distributions per se. The different cryptocurrencies were found to exhibit different volatility structures, with some assets displaying stronger asymmetry, longer memory, or higher sensitivity to shocks than others. This heterogeneity suggests that cryptocurrencies cannot be modeled as a uniform asset class given that they may reflect differences in liquidity, maturity, and investor composition, key factors which influence the performance of the returns of cryptocurrencies.

Chu et al. (2017) therefore positioned IGARCH as a natural benchmark for Bitcoin volatility modeling and highlighted the need for models capable of capturing long persistence and structural heterogeneity. In an indirect response to this study, subsequent research including component GARCH models by Katsiampa (2017), regime-switching GARCH frameworks by Ardia et al. (2019), and smooth-transition approaches as in Cheikh et al. (2020) bring more light to the empirical complexities that Chu et al. (2017) identified.

2.3.3 Component GARCH

In comparing nine GARCH-type models including the standard GARCH, EGARCH, GJR-GARCH, TGARCH, and FIGARCH, Katsiampa (2017) found that the Component GARCH (CGARCH) model offered the best overall fit among these GARCH-family models for cryptocurrencies. CGARCH decomposes volatility into a short-run transitory component and a long-run persistent component, thereby capturing both immediate reactions to market news and deeper structural trends. The finding that Bitcoin's volatility is driven by both short-lived shocks like speculative bursts, exchange incidents, regulatory announcements and slowly evolving forces for example, adoption rates, liquidity growth, macroeconomic uncertainty provides an important conceptual bridge between earlier research work.

In particular, the CGARCH superiority aligns closely with evidence from Chu et al.

(2017), who found the existence extreme persistence and near-unit-root behaviour in cryptocurrencies, often favouring IGARCH specifications. Katsiampa's results refine this view by showing that what appears to be IGARCH-type persistence in single-regime models may actually reflect a combination of long-term variance dynamics and short-term speculative noise. This is a consistent interpretation as was the case in Dyhrberg (2016), who also observed persistent volatility and symmetric responses to shocks, suggesting that Bitcoin's volatility is not purely driven by episodic events but reflects deeper market forces. Thus, Katsiampa's contribution serves as an intermediate step between the early single-regime GARCH evidence and the nonlinear, regime-switching models that followed in later studies.

2.3.4 Regime switching

Ardia et al. (2019) provided evidence that Bitcoin's volatility dynamics are regime-dependent by identifying several low-volatility and high-volatility regimes, highlighting through Markov-switching GARCH models (MSGARCH) that regime switching explains the appearance of IGARCH-like persistence in single-regime models with high-volatility regimes coinciding with market crises and speculative booms. The authors also show that Value-at-Risk forecasts improve under regime-switching models, findings which have motivated later research on nonlinear models and hybrid specifications.

Collectively, these first-generation studies established that Bitcoin exhibits exceptionally high volatility persistence, often approximating IGARCH behaviour, together with largely symmetric responses to return shocks. While the classical GARCH family performed reasonably well in-sample, several limitations became apparent: single-regime models struggled to capture abrupt or gradual shifts in volatility dynamics, and daily data alone appeared insufficient to fully characterise the rapid information flow in crypto markets. These shortcomings motivated a second wave of research that explored asymmetric responses, smooth-transition mechanisms, regime-switching specifications, and high-frequency realised volatility measures to better accommodate the nonlinear and information-rich nature of cryptocurrency volatility.

2.4 Asymmetry, smooth transitions, and realised volatility

Volatility in cryptocurrencies is often asymmetric, persistent, and subject to structural breaks, hence the literature has expanded to include models with nonlinear and smooth-transition dynamics, as well as realised volatility models that exploit high-frequency data.

2.4.1 Asymmetric and smooth-transition dynamics

Cheikh et al. (2020) introduced Smooth Transition GARCH (ST-GARCH) models to analyse gradual, rather than abrupt, shifts between low-volatility and high-volatility states in major cryptocurrencies and found that in the first place, Bitcoin, Ripple and Litecoin portray inverted asymmetry where volatility increases more after positive returns, contrasting strongly with the negative-return leverage effects observed in equities. They also found that volatility responses can vary across cryptocurrencies depending on maturity, liquidity and use base as depicted by Ethereum that exhibits negligible asymmetry and that Bitcoin also exhibited smoother transitions between volatility states than smaller cryptocurrencies which indicates a relatively more stable and liquid market structure.
This is a confirmation of the existence of symmetry in Dyhrberg (2016) and the dominance of symmetric GARCH models in Chu et al. (2017), while extending the literature by showing that asymmetry in crypto markets is not only weak but, in some cases, reversed. The inverted asymmetry pattern supports interpretations of speculation, heavy reliance on momentum-driven trading behaviour, and reinforces the view that volatility in cryptocurrencies is shaped more by behavioural dynamics than by conventional leverage mechanisms.

2.4.2 Realised volatility models

The realised volatility models gained prominence in cryptocurrency research due to their ability to capture intraday information. Corsi (2009) proposed the Heterogeneous

Autoregressive (HAR-RV) model, which approximates long-memory volatility using daily, weekly, and monthly realised volatilities and the subsequent studies applying HAR-RV to Bitcoin found that HAR-RV captures long-memory effects without fractional integration and often, rivals or outperforms GARCH models in out-of-sample forecasts, concluding that high-frequency realised volatility reduces noise and improves forecast accuracy. Later, Dudek et al. (2024) would show further proof that hybrid HAR-ML variants have more improvements on the standalone HAR-RV models.

Although these more sophisticated econometric extensions improved the modeling of nonlinear dynamics and high-frequency information, they still largely treated volatility as a purely statistical phenomenon driven by past returns and realised measures. However, the cryptocurrency market is known to be heavily retail-driven and exceptionally sensitive to online narratives, media coverage, and investor attention. This realisation shifted attention toward behavioural and sentiment-based explanations, leading researchers to investigate whether proxies for investor attention and market sentiment—such as Google search volume, forum discussions, and news tone—could explain volatility patterns beyond what traditional time-series models could capture.

2.5 Behavioural and sentiment drivers

Since the cryptocurrency market is heavily influenced by online narratives, media cycles, social media activities and investor attention, behavioural finance plays an integral role in pricing and valuation of cryptocurrencies. This section, therefore, covers research on forums, news, Google Trends, topic modeling, and sentiment–econometric hybrids.

2.5.1 Social media and forums

Mai et al. (2018) analysed BitcoinTalk forum posts and found that the sentiment from less active "silent majority" users strongly predicts next-day prices, implying that there are "key" investors who directly influence the price of Bitcoin simply by what they say or post online. On the other hand, highly active users or the "vocal minority" influence trading volume, the actual raw numbers of Bitcoin units traded per day, and not the returns. The authors also

show that forum sentiment provides stronger predictive power than Twitter sentiment, a sensible finding given that the '*Bitcointalk*' that they study covers, almost exclusively, news on Bitcoin, unlike Twitter/X on which all sorts of items or news articles are shared.

Thus, Mai and the co-authors highlight the decentralised, retail-driven nature of cryptocurrency markets, where specific community opinions rather than large, influential individuals shape short-term price dynamics. This aligns with later evidence showing that crowd sentiment, search intensity, and collective attention often precede volatility spikes (Wang et al., 2023), providing a behavioural rationale for incorporating sentiment variables into volatility models by distinguishing between market participation intensity and predictive sentiments.

2.5.2 News sentiment

By extracting news articles from major English newspapers, Sapkota (2022) found that mixed positive-negative sentiment has stronger effects on the direction of prices than purely positive or negative sentiment. This is because the mixed positive-negative is a clearer reflection of the real life trading situation. The author also found that sentiment, in general, significantly improves the HAR-RV forecasts, and weekly news is more useful than the daily news in the HAR-RV forecasting framework.

The emphasis on mixed-tone narratives suggests that ambiguity and conflicting news signals generate heightened uncertainty and speculative reactions among investors and the stronger predictive power at the weekly horizon indicates that sentiment diffusion in crypto markets may operate more slowly than in equity markets, possibly due to heterogeneous investor bases and global time-zone effects.

2.5.3 Search-based attention

Search-based measures capture the attention of retail investors and Hoang & Vo (2024) identified that abnormal search volume (ASVI) is integral in predicting next-week crypto-returns, and also influences the increase in volatility and trading volume. ASVI is also shown to have stronger effects on larger cryptocurrencies in comparison to the smaller

units.

This reinforces the attention-based asset pricing literature and justifies the use of Google Trends as a timely behavioural proxy, especially in markets lacking traditional fundamental anchors. The finding that ASVI influences volatility is consistent with the "information-overload" hypothesis, where surges in retail attention precede market disorder and speculative trading. Later ML studies by Wang et al. (2023) incorporate Google Trends as an input feature, showing that attention-based variables enhance nonlinear models more effectively than standard GARCH specifications.

2.5.4 Topic modeling and narrative effects

Coulter (2022) applied Latent Dirichlet Allocation (LDA) to global news coverage and identified as dominant narrative categories crypto-crime, crypto-governance, and crypto-economy. Crypto-crime and governance narratives often preceded sharp daily declines in Bitcoin prices, especially during periods of regulatory uncertainty and exchange hacks, showing that volatility is not only driven by numerical sentiment indices but also by the thematic content and narrative framing of news, a finding consistent with Sapkota (2022) who indicated that sentiment-driven volatility depends not only on tone but also on the type of narrative being amplified.

2.5.5 Hybrid sentiment-econometric models

Hybrid sentiment–econometric models integrate behavioural indicators with traditional time-series structures. For instance, Kabo et al. (2025) showed that sentiment and GDP improve ARIMA and LSTM forecasts, highlighting the complementary value of macroeconomic and behavioural factors and Leber & Egyed (2025) demonstrated that negative sentiment Granger-causes volatility and improves VaR forecasts when incorporated into GARCH–LSTM hybrids.

These studies emphasise that sentiment affects both central tendency and tail risk. They also show that hybrid models can extract nonlinear behavioural effects that GARCH models alone cannot capture, reinforcing the motivation for this study to compare traditional econometric

and machine learning approaches in a sentiment-augmented framework.

2.5.6 Deep learning

Deep learning models such as LSTM, GRU, and CNN architectures outperform traditional econometric models in many studies. Kim & Won (2018) proposed LSTM–GARCH hybrids and subsequently, Wang et al. (2023) found that LSTM models outperform GARCH for weekly and monthly volatility. Eventually, in an attempt to improve accuracy, Han et al. (2025) used GARCH forecasts as LSTM inputs, and the result was a drastic improvement in the prediction accuracy of the LSTM model.

These improvements arise because deep learning architectures capture long-memory patterns, nonlinear relationships, and structural regime changes which are dynamics often persistent in Bitcoin volatility and amplified by sentiment shocks.

2.5.7 Hybrid GARCH – machine learning(ML) models

In an attempt to improve the forecasting powers of the traditional econometric models, researchers have proposed hybrid models, combining the structural interpretability of these econometric models with the adaptability of the machine learning approaches. By assessing the GARCH residual learning, Zahid et al. (2022) showed that ML models learn residual patterns not captured by GARCH. And later on, to incorporate news sentiments and their effects on the model performance, Leber & Egyed (2025) introduced the sentiment-augmented hybrids and reported that these behavioural variables improve performance.

2.5.8 Random Forests for volatility forecasting

While the bulk of early cryptocurrency volatility research relied on extensions of the classical GARCH framework, a parallel strand of the literature has explored machine-learning techniques capable of capturing complex nonlinear patterns and high-dimensional interactions. Among these, Random Forest (RF) regression has emerged as a particularly promising approach for volatility forecasting.

Random Forests, originally proposed by Breiman (2001), are ensemble methods that construct multiple decision trees on bootstrapped samples and aggregate their predictions, thereby reducing variance while maintaining low bias. In the context of volatility modeling, RF offers several advantages over traditional econometric models: it naturally handles nonlinear relationships and interactions among predictors, is robust to multicollinearity and outliers, and provides interpretable measures of feature importance. These properties make RF especially attractive in cryptocurrency markets, where volatility is driven by a mix of past returns, trading volume, and behavioural signals that may interact in highly nonlinear ways.

Empirical applications have demonstrated the potential of RF in financial volatility forecasting. Luong & Dokuchaev (2018) showed that Random Forests outperform the Heterogeneous Autoregressive (HAR) model in squared returns forecasting by better capturing nonlinear interactions among daily, weekly, and monthly realised measures. Basher & Sadorsky (2022) found that RF models outperform logistic regression in predicting Bitcoin price direction, while more recent studies such as Amoanu (2025) and Demirer et al. (2022) reported that RF can outperform traditional econometric specifications when nonlinear predictors or behavioural indices are included. In the specific domain of cryptocurrency volatility, Diane & Brijlal (2024) further illustrated the ability of tree-based ensembles to handle irregular patterns, structural breaks, and sentiment-driven spikes that characterise Bitcoin returns.

The inclusion of sentiment variables further enhances the appeal of Random Forest. Because RF can automatically detect and rank the predictive contribution of lagged Google Trends search interest alongside lagged volatility and volume, it provides a natural benchmark against which to evaluate the incremental value of investor attention within a unified forecasting framework. These features position Random Forest as a strong complement to the classical GARCH family and motivate the direct head-to-head comparison conducted in this thesis.

2.6 Research gap

The expanding literature on sentiment and machine-learning approaches has demonstrated that investor attention and behavioural factors play a first-order role in cryptocurrency volatility and that hybrid and ML models frequently outperform purely econometric specifications. Nevertheless, important gaps remain. Few studies have conducted a systematic head-to-head comparison of classical GARCH-family models (including IGARCH) against machine-learning methods such as Random Forests within a unified forecasting framework. Moreover, sentiment variables are rarely integrated consistently across both econometric and ML approaches on a long, recent sample that encompasses major structural events such as the approval of U.S. spot Bitcoin ETFs.

The above literature review reveals that Bitcoin volatility is extraordinarily persistent, often approximating IGARCH behaviour, and that volatility dynamics are nonlinear, asymmetric, and regime-dependent. Sentiment, captured through search activity, news tone, and online forums, has emerged as a major driver of volatility. Yet despite these insights, a systematic comparison of GARCH-family models with Random Forests that explicitly incorporates a sentiment proxy on an extended post-2017 sample remains scarce.

This thesis addresses these gaps by conducting a sentiment-augmented comparative analysis of GARCH and Random Forest models using daily Bitcoin data from January 2017 to September 2025. While a substantial body of cryptocurrency volatility research has appeared in specialised or lower-tier outlets, the core empirical patterns documented in this literature—particularly the extremely high persistence of volatility, limited asymmetry, and the dominant role of lagged variance—are consistently reproduced in the present analysis. The current study therefore builds upon and extends this body of work by employing a longer and more recent sample period (2017–2025) that encompasses major structural developments, most notably the approval of U.S. spot Bitcoin ETFs in 2024. This allows for a methodologically rigorous comparison of classical GARCH-family models with machine-learning approaches, particularly Random Forests used in this study, within a unified forecasting framework, while explicitly incorporating a sentiment proxy based on Google Trends search interest.

Chapter 3: Methodology

3.1 Introduction

This chapter describes the econometric and machine-learning methodology used to forecast Bitcoin volatility and assesses whether sentiment-enhanced nonlinear methods outperform standard GARCH benchmarks. The chapter specifies the data-generating process for returns, the conditional variance dynamics under GARCH-type models, the Random Forest forecasting set-up, the out-of-sample forecasting design, and the forecast evaluation framework and multiple-model comparison procedures.

3.2 Notation and return construction

Let P_t denote the daily closing price of Bitcoin on day t. The continuously compounded (log) return is defined as

$$r_t = 100\,(\ln P_t - \ln P_{t-1}) \tag{3.1}$$

where the scaling factor of 100 expresses returns in percentage terms. Log returns are the standard choice in financial econometrics because they are additive over time, approximately symmetric, and possess convenient statistical properties for volatility modeling (Bollerslev, 1986).

Let $\mathcal{F}_{t-1}$ denote the information set available at time $t-1$. Returns are modelled as conditionally heteroskedastic processes, i.e., the conditional mean may follow a simple autoregressive structure while the conditional variance σ_t^2 is time-varying. A general conditional mean specification is

$$r_t = \mu + \sum_{i=1}^{p} \phi_i r_{t-i} + \varepsilon_t, \tag{3.2}$$

where the innovation satisfies

$$\mathbb{E}(\varepsilon_t \mid \mathcal{F}_{t-1}) = 0, \qquad \mathbb{V}(\varepsilon_t \mid \mathcal{F}_{t-1}) = \sigma_t^2. \tag{3.3}$$

In line with the vast majority of volatility studies, the present analysis focuses primarily on modeling and forecasting the conditional variance σ_t^2. The conditional mean is therefore treated as a secondary component (often a simple constant or low-order AR process) because the main interest lies in the dynamics of volatility rather than in predicting the expected return itself.

3.2.1 Investor attention proxy

Daily worldwide search interest for the keyword "Bitcoin" was collected from Google Trends using the `pytrends` Python library (worldwide web search, English language). Because Google Trends imposes length restrictions on individual daily queries, the data were retrieved in successive non-overlapping six-month windows from 1 January 2017 to the end of the sample period. Each window is independently normalized by Google Trends to the range 0–100, where 100 represents the peak search interest within that specific window. The resulting files were concatenated and the full series was re-normalized to a consistent 0–100 scale using the global minimum and maximum search interest over the entire sample. The final variable `Search_Interest` therefore measures *relative* daily investor attention to Bitcoin and is lagged by one day (S_{t-1}) when entering the conditional variance equation of the GARCH-X and IGARCH-X models.

3.3 Volatility proxy

Since true conditional volatility is a latent variable that cannot be observed directly, all forecast evaluation in this study is based on a proxy constructed from daily returns. The

primary proxy employed is the squared daily log return

$$RV_t = r_t^2, \tag{3.4}$$

where $r_t = 100 \times \ln(\frac{P_t}{P_{t-1}})$ denotes the continuously compounded percentage return. This measure is a standard and widely used approximation in daily-frequency volatility studies when high-frequency intraday data are unavailable (Patton, 2011; Bollerslev, 1986).

It is important to note that this is *not* a realised volatility measure in the conventional high-frequency sense, that is to say, the sum of squared intraday returns over short intervals such as 5 minutes as true realised volatility typically requires intraday price data. The squared daily return used here is therefore more accurately described as a simple daily volatility proxy. Although this proxy is known to be noisy, it remains the conventional choice in the GARCH literature at daily frequency and allows direct comparability with the majority of existing cryptocurrency volatility studies.

For robustness, alternative proxies based on rolling windows of squared returns are also considered

$$RV_t^{(h)} = \frac{1}{h} \sum_{j=0}^{h-1} r_{t-j}^2, \tag{3.5}$$

for horizons $h \in \{5, 10, 20\}$. These checks help assess whether the main conclusions are sensitive to the specific choice of volatility proxy.

3.4 Econometric volatility models

The empirical analysis in this thesis is built around four core econometric specifications: the standard GARCH(1,1), the Integrated GARCH (IGARCH(1,1)), and their sentiment-augmented counterparts (GARCH-X and IGARCH-X). These models were selected because they directly address the stylised facts identified in the literature review—particularly the extreme volatility persistence and near-unit-root behaviour frequently documented in Bitcoin returns—while remaining parsimonious and easily interpretable. This section presents the model specifications, their theoretical properties, and the rationale for their use in the cryptocurrency context.

3.4.1 Shock decomposition and innovation distribution

All models decompose the return innovation as

$$\varepsilon_t = \sigma_t z_t, \tag{3.6}$$

where $\{z_t\}$ is an i.i.d. sequence with $\mathbb{E}(z_t) = 0$ and $\mathbb{V}(z_t) = 1$. To accommodate the heavy tails and excess kurtosis commonly observed in cryptocurrency returns (Chu et al., 2017; Katsiampa, 2017), estimation is performed under two alternative distributional assumptions for the standardised innovation z_t

$$z_t \sim \mathcal{N}(0,1) \quad \text{or} \quad z_t \sim t_\nu \quad \text{(scaled to unit variance)}. \tag{3.7}$$

The Student-t distribution is particularly relevant for Bitcoin, while the normal distribution serves as the classical benchmark. Both specifications are estimated and compared throughout the analysis.

3.4.2 GARCH(1,1)

The baseline specification is the classic GARCH(1,1) model of Bollerslev (1986):

$$\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2, \tag{3.8}$$

subject to the positivity constraints

$$\omega > 0, \qquad \alpha \geq 0, \qquad \beta \geq 0. \tag{3.9}$$

Covariance stationarity requires $\alpha + \beta < 1$, in which case the unconditional variance is

$$\mathbb{E}(\sigma_t^2) = \frac{\omega}{1 - \alpha - \beta}. \tag{3.10}$$

The sum $\alpha + \beta$ directly measures volatility persistence. Values close to one (as typically found in Bitcoin) imply that shocks decay very slowly. The GARCH(1,1) therefore serves

as the natural benchmark because of its parsimony and strong empirical performance across asset classes.

3.4.3 IGARCH(1,1)

Given the near-unit-root behaviour frequently documented in cryptocurrency volatility, the Integrated GARCH (IGARCH(1,1)) specification is also estimated. It imposes the restriction

$$\alpha + \beta = 1, \tag{3.11}$$

yielding

$$\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + (1-\alpha)\sigma_{t-1}^2. \tag{3.12}$$

Under IGARCH, volatility shocks do not decay over time but persist indefinitely. This formulation is particularly relevant for Bitcoin, where large shocks (regulatory announcements, exchange hacks, ETF approvals) have been shown to exert long-lasting effects on market uncertainty.

3.4.4 Sentiment-augmented variance dynamics (GARCH-X and IGARCH-X)

To examine whether investor attention contains incremental predictive power beyond lagged returns and volatility, Google Trends search interest (S_{t-1}) is introduced as an exogenous regressor in the variance equation. The sentiment-augmented specification is

$$\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2 + \gamma S_{t-1}, \tag{3.13}$$

where the coefficient γ measures the marginal contribution of attention shocks to next-period conditional variance. Both the standard GARCH and the IGARCH versions are estimated with and without the sentiment term, producing the four core models used in the empirical analysis. This extension is directly motivated by the behavioural literature reviewed in Chapter 2, which consistently shows that retail attention is a powerful driver of

cryptocurrency volatility. The lagged structure ensures that the information is available at the time of forecasting, preserving the real-time nature of the exercise.

By comparing the four specifications under both normal and Student-t innovations, the analysis provides a comprehensive assessment of how well classical econometric models capture Bitcoin volatility and whether readily available sentiment data improves forecast accuracy.

3.5 Estimation via maximum likelihood

All four econometric models (GARCH, IGARCH, GARCH-X, and IGARCH-X) are estimated by maximum likelihood. This approach is standard in the GARCH literature because it yields consistent and asymptotically efficient parameter estimates under mild regularity conditions and naturally accommodates alternative innovation distributions (Bollerslev, 1986).

Let θ denote the vector of parameters, which includes the conditional variance parameters ω, α, β (and γ in the sentiment-augmented models) together with any distributional parameters such as the degrees of freedom ν for the Student-t distribution. Conditional on the information set $\mathcal{F}_{t-1}$, the log-likelihood contribution of observation t is given by

$$\ell_t(\theta) = \log f(z_t(\theta)) - \frac{1}{2}\log(\sigma_t^2(\theta)), \qquad z_t(\theta) = \frac{\varepsilon_t}{\sigma_t(\theta)}, \tag{3.14}$$

where $f(\cdot)$ denotes the density of the standardised innovation z_t. The sample log-likelihood is then the sum of the individual contributions

$$\mathcal{L}(\theta) = \sum_{t=1}^{T} \ell_t(\theta). \tag{3.15}$$

The parameter vector $\widehat{\theta}$ is obtained by maximising $\mathcal{L}(\theta)$ subject to the positivity and stationarity constraints discussed in section 3.4

Maximum likelihood estimation is particularly convenient in this setting because it allows direct comparison of models with different innovation distributions (normal versus

Student-t) through standard information criteria such as AIC and BIC. It also provides the asymptotic standard errors needed for inference on the persistence parameters and the sentiment coefficient γ. The resulting parameter estimates form the basis for the one-step-ahead variance forecasts evaluated in Chapter 4.

3.6 Forecast definition

A central objective of this thesis is to evaluate the out-of-sample predictive ability of the competing models. To mimic the real-time environment faced by a forecaster, all models are estimated recursively using a fixed-length rolling window and then used to produce one-step-ahead volatility forecasts. This walk-forward (or recursive) forecasting scheme ensures that forecasts are genuinely out-of-sample and that parameter estimates are based only on information available up to the forecast origin.

For any estimated GARCH-type model, the one-step-ahead conditional variance forecast at time t is

$$\widehat{\sigma}^2_{t+1|t} = \widehat{\omega} + \widehat{\alpha}\widehat{\varepsilon}^2_t + \widehat{\beta}\widehat{\sigma}^2_t \quad \text{plus } \widehat{\gamma} S_t \text{ for the sentiment-augmented models.} \tag{3.16}$$

In the sentiment-augmented specifications (GARCH-X and IGARCH-X), the lagged Google Trends search interest S_t is known at the forecast origin and is therefore included directly in the variance equation. For the baseline GARCH and IGARCH models, the sentiment term is simply omitted.

The forecasts are generated using an expanding-window or fixed rolling-window design. In the main analysis a fixed rolling window of 750 trading days is employed (approximately three years of data), which strikes a balance between statistical precision and the ability to adapt to changing market conditions. At each step t, the model is re-estimated using only the most recent 750 observations available up to time t, and the one-step-ahead forecast for day $t + 1$ is produced. The estimation window is then rolled forward by one observation and the process is repeated. This recursive procedure yields a complete sequence of genuine out-of-sample forecasts for the evaluation period (November 2023 to September 2025).

This forecasting design is deliberately aligned with the implementation in the R estimation routine described in Chapter 4 and ensures full comparability between the econometric models and the Random Forest forecasts. It also reflects realistic practitioner behaviour, where models are updated frequently to incorporate the latest market information. The resulting one-step-ahead variance forecasts form the basis for all subsequent forecast evaluation exercises (QLIKE loss, SPA test, MCS procedure, and robustness checks) reported in Chapter 4.

3.7 Random Forest forecasting set-up

While the GARCH-family models impose a specific parametric structure on the conditional variance, the Random Forest (RF) approach offers a flexible, nonparametric alternative that can capture complex nonlinear interactions and high-dimensional relationships without strong functional-form assumptions. This section describes the Random Forest regression framework used in this thesis and its implementation for one-step-ahead Bitcoin volatility forecasting.

Random Forests, introduced by Breiman (2001), are an ensemble learning method that constructs a large number of regression trees on bootstrapped samples of the training data and aggregates their predictions. Each tree is grown by recursively partitioning the feature space so as to minimise the mean squared error at every split. At each node, a random subset of features is considered for splitting, which decorrelates the individual trees and reduces the variance of the ensemble. The final forecast is the simple average of the predictions from all trees

$$\widehat{RV}_{t+1|t} = \frac{1}{B}\sum_{b=1}^{B} T_b(\mathbf{X}_t), \tag{3.17}$$

where B is the number of trees, $T_b(\cdot)$ denotes the prediction of the b-th tree, and $\mathbf{X}_t$ is the vector of predictors available at time t.

In this thesis, the target variable is the squared returns $RV_{t+1} = r_{t+1}^2$ (squared log returns), consistent with the GARCH-family models. The feature vector $\mathbf{X}_t$ is constructed to capture both ARCH-style dynamics and behavioural information and includes the first and fifth lags

of squared returns, the lagged absolute return, the lagged log return, lagged log trading volume, and both the first and fifth lags of Google Trends search interest. These predictors mirror the information set available to the econometric models while allowing the Random Forest to automatically detect nonlinear interactions and threshold effects that a linear specification cannot capture.

To avoid look-ahead bias and respect the time-series nature of financial data, model selection and hyperparameter tuning are performed using time-series cross-validation (TimeSeriesSplit). The number of trees ($n_estimators$) and maximum tree depth are chosen via grid search within each training window. The final Random Forest is then used to produce genuine one-step-ahead forecasts that are perfectly aligned with the GARCH-family forecasts in both time and evaluation sample. This alignment ensures that all ten models, that is eight econometric specifications plus two Random Forest variants are evaluated on exactly the same out-of-sample period using identical squared returns proxies.

The Random Forest specification therefore serves two important purposes. First, it provides a strong nonparametric benchmark against which the parametric GARCH models can be judged. Second, because it can naturally incorporate the lagged Google Trends variable alongside traditional volatility and volume lags, it offers a clean test of whether investor attention adds predictive value beyond what is already captured by the classical autoregressive structure. The full implementation details, including the exact feature set and hyperparameter grid, are provided in the accompanying Python notebook.

3.8 Forecast evaluation - loss functions and model comparison

A central challenge in volatility forecasting is that the true conditional variance is latent and can only be proxied by a noisy realised measure (squared returns in this study). Simple loss functions such as mean squared error (MSE), root mean squared error (RMSE), or mean absolute error (MAE) are sensitive to noise in the proxy and can produce misleading rankings. To overcome these limitations, the evaluation framework in this thesis follows the

modern best-practice approach of Patton (2011) and employs three complementary tools: the QLIKE loss function, the Superior Predictive Ability (SPA) test, and the Model Confidence Set (MCS) procedure. These methods are specifically designed to be robust to proxy noise and to control for data-snooping when multiple models are compared.

3.8.1 QLIKE loss

The primary evaluation criterion is the QLIKE loss function. For any model m, the loss at time $t+1$ is

$$L_{t+1}(m) = \frac{RV_{t+1}}{\widehat{\sigma}^2_{t+1|t}(m)} - \ln\left(\frac{RV_{t+1}}{\widehat{\sigma}^2_{t+1|t}(m)}\right) - 1, \tag{3.18}$$

where RV_{t+1} is the squared returns and $\widehat{\sigma}^2_{t+1|t}(m)$ is the one-step-ahead forecast.

As shown by Patton (2011), QLIKE is the only loss function that remains consistent for volatility ranking even when the proxy RV_{t+1} is noisy, provided the true loss is a function of variance. In contrast, MSE and RMSE can produce biased model rankings because they penalise large forecast errors quadratically and are highly sensitive to extreme realisations in the proxy. QLIKE therefore provides a more reliable basis for comparing the ten competing models (eight GARCH-family specifications and two Random Forest variants) and is used as the central loss function for all ranking, SPA, and MCS procedures.

3.8.2 Loss differentials and benchmark comparison

To quantify the economic magnitude of differences in forecast accuracy, loss differentials relative to the benchmark model (GARCH-Normal) are computed. Let model 0 denote the benchmark. The loss differential for any competing model m at time $t+1$ is defined as

$$d_{t+1}(m) = L_{t+1}(0) - L_{t+1}(m), \tag{3.19}$$

where $L_{t+1}(\cdot)$ is the QLIKE loss. A positive value of $d_{t+1}(m)$ indicates that model m produces, on average, a lower loss (i.e., more accurate forecasts) than the benchmark under the chosen loss function.

These differentials serve two important purposes. First, they provide an intuitive and

economically interpretable measure of forecast improvement that can be averaged over the entire out-of-sample period to assess practical significance. Second, they form the fundamental building blocks for the formal statistical tests used in this study—the Superior Predictive Ability (SPA) test and the Model Confidence Set (MCS) procedure. By transforming the raw loss series into differentials, the analysis directly quantifies how much better (or worse) each of the nine competing models performs relative to the GARCH-Normal benchmark, thereby complementing the QLIKE ranking with a clearer picture of economic relevance.

3.8.3 Superior predictive ability (SPA) test

The Superior Predictive Ability (SPA) test of Hansen (2005) formally evaluates whether any competing model significantly outperforms the benchmark while explicitly controlling for the data-snooping bias that arises when many models are compared simultaneously. The test statistic is based on the maximum mean loss differential:

$$T^{\text{SPA}} = \max_{m \in \mathcal{M}} \sqrt{T} \cdot \overline{d}(m), \tag{3.20}$$

where $\overline{d}(m)$ is the sample average of the loss differentials. The null hypothesis states that no model is superior to the benchmark

$$H_0 : \max_{m \in \mathcal{M}} \mathbb{E}\left[d_t(m)\right] \leq 0. \tag{3.21}$$

Inference is conducted via a block bootstrap that preserves the serial dependence present in the loss differentials. A high p-value (as obtained in the empirical analysis) indicates that none of the alternative specifications offers statistically significant superior predictive performance.

3.8.4 Model confidence set (MCS)

To compare the full set of ten models simultaneously, the Model Confidence Set (MCS) procedure of Hansen et al. (2011) is applied. The MCS identifies a subset $\widehat{\mathcal{M}}_{1-\alpha}$ of models

that are statistically indistinguishable from the best model at confidence level $1-\alpha$ (typically 90%). At each iteration, the range statistic

$$T_R = \max_{i,j \in \mathcal{M}} \left| \overline{d}_{ij} \right| \tag{3.22}$$

is computed, where $\overline{d}_{ij}$ is the average loss differential between models i and j. Models that are found to be significantly inferior are eliminated until the null of equal predictive ability can no longer be rejected within the remaining set. The final MCS therefore contains all models that cannot be statistically ruled out as belonging to the set of superior forecasters. Together, the QLIKE loss, loss differentials, SPA test, and MCS procedure provide a statistically rigorous and proxy-robust framework for evaluating and ranking the forecasting performance of the competing models. The empirical results of this evaluation are presented and discussed in Chapter 4.

Chapter 4: Empirical Results

4.1 Preliminary data analysis

4.1.1 Core distributional statistics

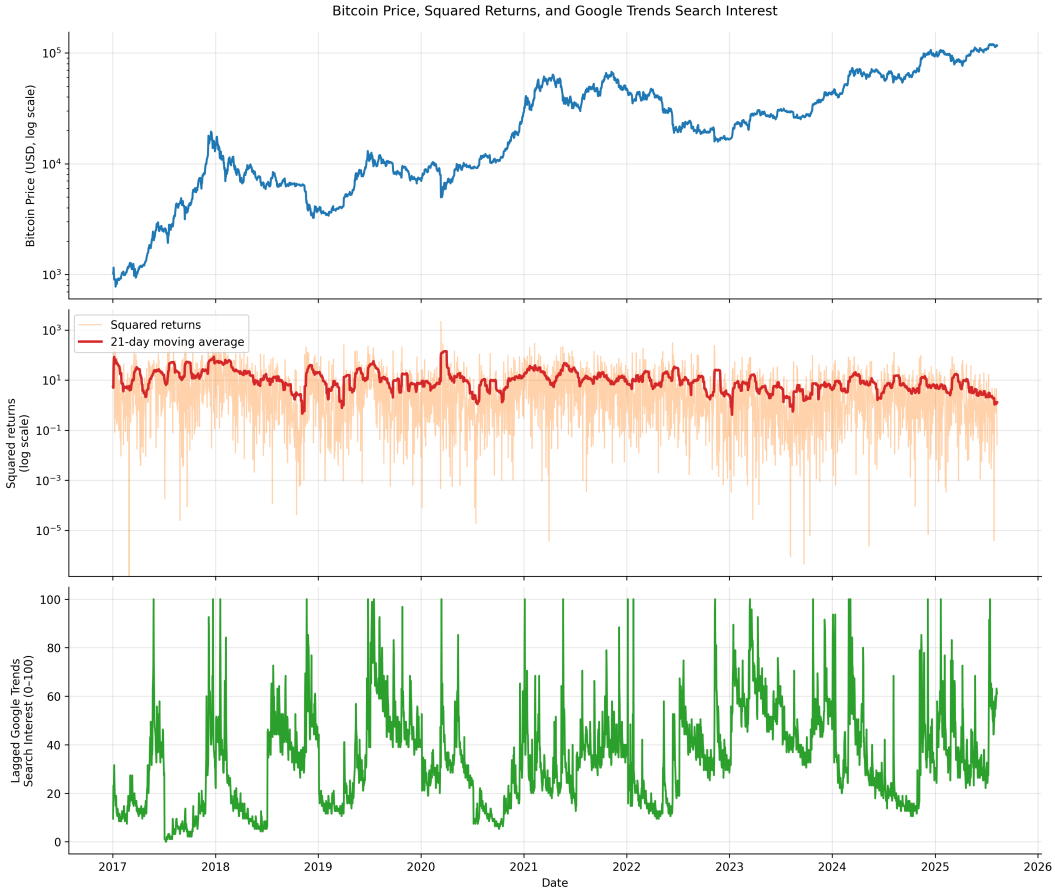


Figure 4.1: *Time series of Bitcoin price (log scale), squared returns, and lagged Google Trends search interest (0–100 scale).*

Table 4.1 presents the core distributional statistics for Bitcoin log returns, squared returns, and lagged Google Trends search interest over the period 2017–2025, and figure 4.1 provides the visual counterpart, showing the evolution of these three series. The Bitcoin price series (top panel) exhibits strong upward trends during the major bull markets of 2017,

Table 4.1: *Core descriptive statistics for returns, squared returns, and lagged Google Trends search interest.*

	Returns	Squared Returns	Lagged Sentiment
Mean	0.1515	13.5990	34.2722
Median	0.1275	2.3597	32.6316
Std. Dev.	3.6852	49.9664	19.3886
Min	-46.4730	0	0
Max	22.5119	2159.7414	100
Skewness	-0.6597	26.9230	0.6589
Excess Kurtosis	11.6638	1093.4381	0.3205

2020–2021, and 2024–2025, interspersed with pronounced drawdowns. The corresponding log returns which are not shown directly but used to construct the middle panel, have a modest positive mean of 0.1515% per day but display substantial variability, with a standard deviation of 3.6852%. This high volatility is clearly visible in the middle panel, where squared returns series shows strong clustering and occasional extreme spikes — most notably during the March 2020 COVID-19 crash. The squared returns series has a mean of 13.5990 and an extremely high standard deviation of 49.9664, reflecting the presence of very large but infrequent volatility outbursts. The bottom panel shows lagged Google Trends search interest, which fluctuates between 0 and 100 with an average of 37.56 and a standard deviation of 18.42, indicating moderate but persistent variation in investor attention.

Higher-order moments further confirm the stylised facts of cryptocurrency returns. In this regard the log returns exhibit negative skewness (-0.6597) and very high excess kurtosis (11.6638), consistent with the frequent occurrence of large negative price movements and fat tails. The squared returns displays even more extreme characteristics, with massive positive skewness (26.9230) and extraordinarily high excess kurtosis (1093.4381), driven by the occasional but very large volatility spikes visible in the middle panel. In contrast, the sentiment series is comparatively well-behaved, showing mild positive skewness (0.6589) and low excess kurtosis (0.3205). This difference suggests that the relationship between investor attention and volatility is unlikely to be purely linear and may instead operate through nonlinear or threshold effects.

Overall, the visual evidence in figure 4.1 and the summary statistics in table 4.1 jointly

highlight the extreme volatility, strong persistence, and fat-tailed nature of Bitcoin returns, while also revealing substantial time variation in investor attention, all of which motivate the econometric and machine-learning models employed in the subsequent analysis.

4.1.2 Bitcoin price dynamics

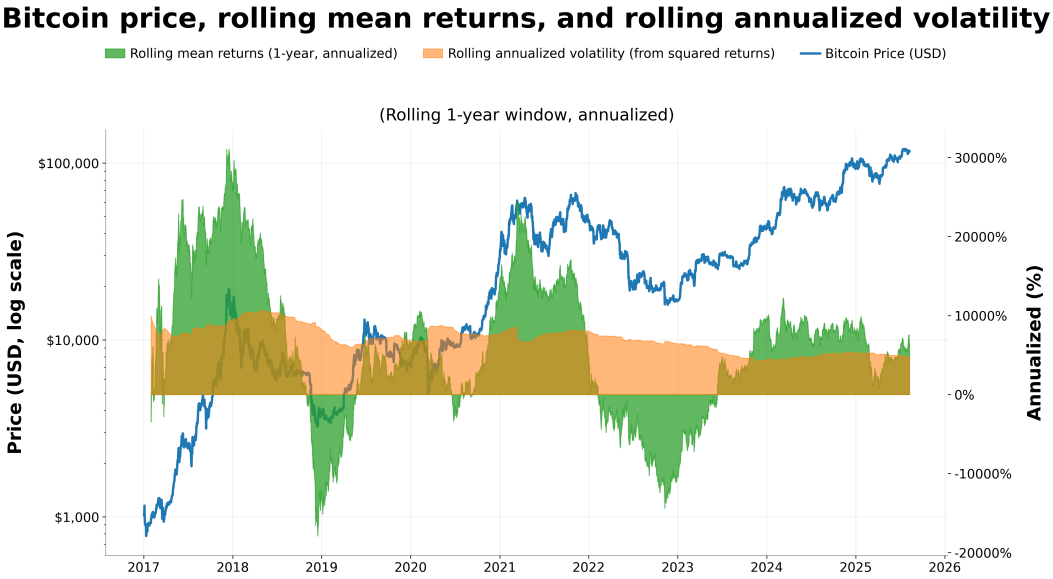


Figure 4.2: *Bitcoin price dynamics, rolling mean returns and squared returns*

Figure 4.2 illustrates the relationship between Bitcoin price dynamics, rolling mean returns, and squared returns using one-year rolling windows, combining three elements: the Bitcoin price series (left axis, logarithmic scale), the annualised rolling mean returns, and the annualised squared returns computed from squared daily returns on the right axis. It reveals that periods of strong price appreciation coincide with elevated average returns and increased volatility, a pattern that is particularly visible during the major bull market cycles around 2017–2018 and 2020–2021, where large positive rolling returns are accompanied by pronounced spikes in squared returns, reflecting the speculative nature of these episodes, periods during which rapid price increases are associated with heightened uncertainty and large price fluctuations.

The figure also shows that volatility exhibits clear clustering over time as periods of high volatility tend to persist for extended intervals rather than fluctuate randomly before gradually declining. These clusters correspond closely with major market events and regime

shifts in the Bitcoin market, such as the boom–bust cycles observed in 2018 and the subsequent recovery in 2020–2021.

The series also shows a substantial variation in the magnitude of rolling mean returns across market phases. During bullish periods, the annualised rolling returns become strongly positive as evident between 2018 - 2019 and 2020 - 2021, while prolonged bearish phases generate negative rolling returns as in 2019 and 2022 - 2023. In contrast, the squared returns remains strictly positive and typically declines during periods of relative market stability.

4.1.3 Quantiles and tail behaviour

Table 4.2: *Quantiles illustrating tail behaviour of returns, squared returns, and lagged sentiment.*

	Returns	Squared Returns	Lagged Sentiment
1%	-10.6030	0.0007	2.5368
5%	-5.8732	0.0128	7.3684
25%	-1.3578	0.3872	17.8947
50%	0.1275	2.3597	32.6316
75%	1.7339	10.7193	46.3158
95%	5.8442	59.4999	68.4211
99%	10.1557	146.3326	92.6316

The median daily return reported in table 4.2 is approximately 0.1275%, which is close to zero and consistent with the near-zero unconditional mean which is typically observed in financial return series. However, the tails show mild asymmetry with the 1% quantile of returns being -10.6030%, slightly exceeding the magnitude of the 99% quantile of 10.1557%, suggesting a somewhat greater downside risk, which aligns with the negative skewness reported in the descriptive statistics in table 4.1.

The volatility dynamics become substantially more pronounced when examining squared returns given that while the median squared return is relatively small at 2.3597, the upper tail expands rapidly as the 95% quantile reaches 59.4999 and the 99% quantile increases sharply to 146.3326, indicating that volatility escalates dramatically during extreme market episodes. On the other hand, the lower tail remains close to zero, with a 1% quantile of 0.0007, which depicts prolonged periods of relatively calm market conditions punctuated by

rare but intense volatility spikes.

The quantile range for the sentiment variable lies between 2.5368 at the 1% level and 92.636 at the 99% level, with a median value of 32.6316, which is a wide dispersion indicating that investor attention fluctuates substantially over time. However, unlike the volatility proxy, the sentiment distribution does not exhibit explosive tail behaviour, owing to the 0–100 scaling inherent in Google Trends data.

The quantile evidence, therefore, reinforces the presence of heavy-tailed behaviour in Bitcoin returns and especially in volatility measures and while typical market fluctuations are captured within the interquartile range (0.3872 to 10.7193 for the squared returns), the rapid escalation between the 95% and 99% quantiles highlights the severity of extreme volatility episodes that characterise cryptocurrency markets.

4.2 Dependence and persistence diagnostics

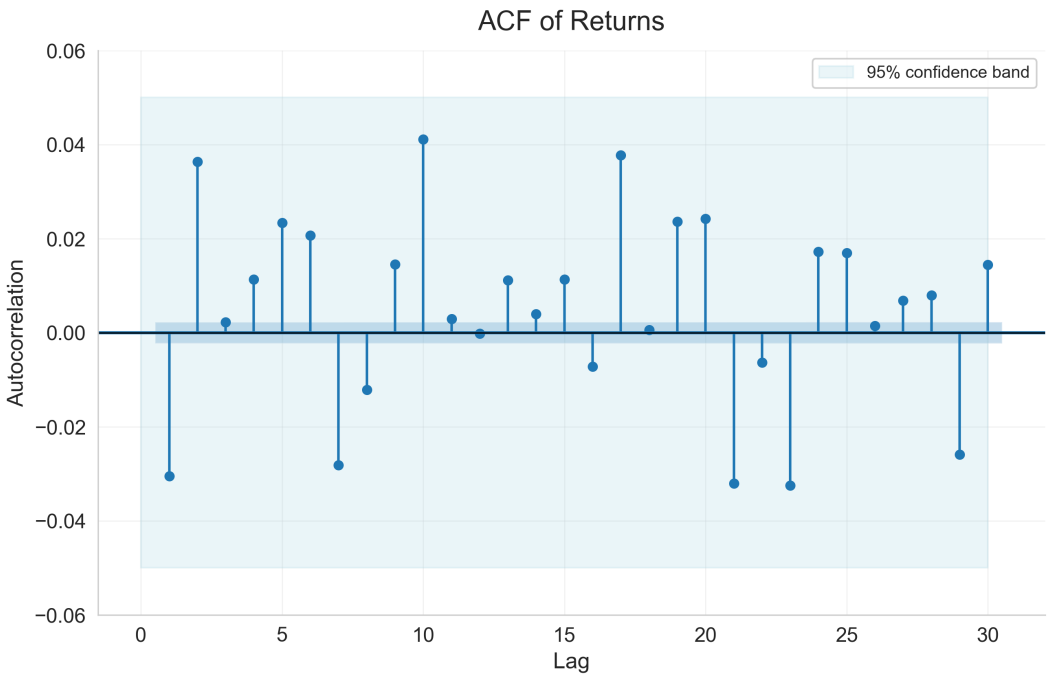


Figure 4.3: *ACF of returns*

The autocorrelation function of returns in Figure 4.3 shows that most autocorrelations lie within the 95% confidence bands and decay rapidly, indicating that serial dependence in the

conditional mean is generally weak and short-lived. However, a few isolated spikes at lower lags suggest the presence of limited short-run linear dependence.

This is broadly consistent with the Ljung–Box test results reported in table 4.3. At lag 10, the null hypothesis of no autocorrelation is rejected at the 5% level (p-value = 0.0344), indicating some short-term dependence in returns. However, at longer horizons (lags 20 and 30), the p-values increase to 0.0948 and 0.0981, respectively, implying that the null cannot be rejected at conventional significance levels.

Table 4.3: *Ljung–Box test p-values for dependence diagnostics of returns and squared returns.*

	Returns	Squared Returns
Lag 10	0.0344	0.0000
Lag 20	0.0948	0.0000
Lag 30	0.0981	0.0000

In contrast, the autocorrelation function of squared returns in figure 4.4 displays strong and persistent positive autocorrelation. The first few lags exhibit relatively high autocorrelation, followed by a slow and gradual decay that remains positive over many lag horizons, a pattern characteristic of pronounced volatility clustering, whereby periods of high volatility tend to be followed by further periods of elevated volatility.

The Ljung–Box test results in table 4.3 strongly support this observation as the p-values for squared returns are effectively zero across all reported lags, leading to a clear rejection of the null hypothesis of no serial dependence.

These results indicate that while there is some evidence of short-run autocorrelation in raw returns, particularly at lower lags, such dependence is weak and not persistent. On the other hand, the strong and persistent autocorrelation observed in squared returns provides clear evidence of volatility clustering and conditional heteroskedasticity in the return series. These findings, therefore, suggest that while the conditional mean is largely unpredictable, the conditional variance exhibits significant persistence, thereby justifying the use of ARCH/GARCH-type models to capture the dynamics of volatility.

These tests reveal that while returns themselves display minimal autocorrelation, the volatility proxy exhibits strong persistence over time which provides a direct empirical

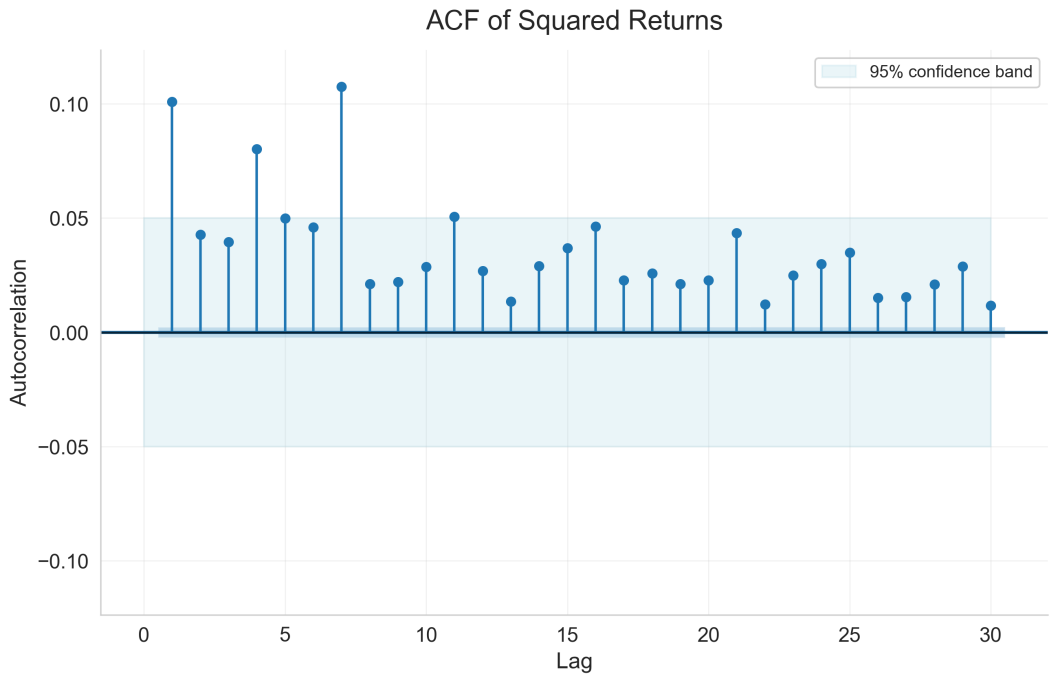


Figure 4.4: *ACF of squared returns*

justification for the use of ARCH-type models designed to capture time-varying conditional heteroskedasticity. Furthermore, the slow decay in the autocorrelation of squared returns suggests highly persistent variance dynamics, motivating the comparison between stationary GARCH(1,1) and near-integrated IGARCH specifications in the subsequent modeling section.

4.3 Distributional and stationarity diagnostics

4.3.1 Normality tests

Table 4.4: *Jarque–Bera normality test results for returns and squared returns.*

	Returns	Squared Returns
Observations	3142	3142
JB Statistic	17975.6826	156405979.3164
p-value (JB)	0.0000	0.0000

The Jarque–Bera (JB) statistics reported in table 4.4 strongly reject the null hypothesis

of normally distributed data for both returns and the squared returns with a JB statistic of 17 975.68 for the returns with a p-value of 0.0000, indicating significant departures from normality and the deviation is even more pronounced for squared returns, where the JB statistic exceeds 156 million, reflecting extremely high skewness and kurtosis. These further confirm the heavy-tailed nature of Bitcoin return dynamics and the extreme right-skewness present in volatility proxies which imply that extreme observations are more frequent in occurance than would be predicted under a standard Gaussian distribution. Consequently, assuming normally distributed innovations would underestimate the probability of large market movements which motivates the use of heavy-tailed innovation distributions, such as the Student-t, in the estimation of GARCH-type models.

While the Jarque–Bera results clearly indicate strong departures from normality in the unconditional distribution of returns and volatility proxies, this does not imply that heavy-tailed innovation distributions will necessarily improve forecasting performance in GARCH-type models as the JB test captures unconditional properties of the data, whereas GARCH models operate in a conditional framework. Once volatility clustering is accounted for, much of the excess kurtosis is absorbed by the variance dynamics, meaning that the standardized residuals need not exhibit the same degree of non-normality as the raw series. Moreover, although Student-t innovations improve in-sample fit, as reflected in higher likelihood values and lower information criteria, this additional flexibility can come at the cost of poorer out-of-sample performance.

4.3.2 Stationarity diagnostics

Table 4.5: *ADF unit root test results for returns, squared returns, and lagged sentiment.*

	Returns	Squared Returns	Lagged Sentiment
ADF Statistic	-38.8121	-16.7408	-4.6501
p-value	0.0000	0.0000	0.0001
Used Lags	1	6	29
Observations	3140	3135	3112
Critical Value (5%)	-2.8625	-2.8625	-2.8625

The ADF test results in table 4.5 reject the null hypothesis of a unit root for all variables at conventional significance levels where the returns display particularly strong evidence of stationarity with an ADF statistic of -38.8121 and a p-value of 0.0000. Similarly, the volatility proxy and the sentiment variable also reject the unit root null with ADF statistics of -16.7408 and -4.6501 respectively and p-values well below 0.05 in both cases. This indicates that all series are stationary in levels and do not require differencing before modeling.

Table 4.6: *KPSS stationarity test results for returns, squared returns, and lagged sentiment.*

	Returns	Squared Returns	Lagged Sentiment
Observations	3142	3142	3142
KPSS Statistic	0.1430	2.1518	1.2838
p-value	0.1000	0.0100	0.0100
Used Lags	8	17	31

To provide a complementary assessment of stationarity, the KPSS test results reported in table 4.6 show that for the returns series, the KPSS statistic is 0.1430 with a p-value of 0.1000, indicating that the null hypothesis of stationarity cannot be rejected while the KPSS statistics for squared returns (2.1518) and lagged sentiment (1.2838) are associated with p-values of 0.0100, leading to rejection of the stationarity null for these variables.

The divergence between the ADF and KPSS results suggests that while volatility and sentiment do not contain unit roots, they may exhibit structural shifts or evolving regimes over the sample period which may further prove the presence of volatility clustering and changing market conditions in cryptocurrency markets.

4.3.3 Correlation analysis

As shown in table 4.7, returns display a modest negative correlation with squared returns (-0.1569), meaning that negative price movements tend to coincide with higher volatility, while the absolute returns and squared returns, on the other hand, exhibit a strong positive correlation (0.7690), which confirms that both variables capture closely related information about the magnitude of price fluctuations and both measures behave in a similar manner as volatility proxies, although squared returns amplify extreme movements relative to absolute

Table 4.7: *Pearson correlation matrix for returns, absolute returns, squared returns, and lagged sentiment.*

	Returns	Absolute Returns	Squared Returns	Lagged Sentiment
Returns	1.0000	-0.0326	-0.1569	-0.0197
Absolute Returns	-0.0326	1.0000	0.7690	0.0213
Squared Returns	-0.1569	0.7690	1.0000	0.0040
Lagged Sentiment	-0.0197	0.0213	0.0040	1.0000

returns.

The sentiment variable shows only very weak contemporaneous correlations with the financial variables, with a negative correlation with the returns(-0.0197), while correlations with absolute returns (0.0213) and squared returns (0.0040) are close to zero, meaning that Google Trends as a sentiment variable does not exhibit a strong linear relationship with Bitcoin price movements or volatility.

Table 4.8: *Spearman rank correlation matrix for returns, absolute returns, squared returns, and lagged sentiment.*

	Returns	Absolute Returns	Squared Returns	Lagged Sentiment
Returns	1.0000	0.0754	0.0754	-0.0335
Absolute Returns	0.0754	1.0000	1.0000	0.0030
Squared Returns	0.0754	1.0000	1.0000	0.0030
Lagged Sentiment	-0.0335	0.0030	0.0030	1.0000

To provide a non-parametric measure of association that is less sensitive to extreme observations, table 4.8 presents the Spearman rank correlation matrix, which displays broadly similar patterns to the Pearson correlations in table 4.7. The return series maintains weak correlations with volatility measures, while absolute returns and squared returns remain perfectly aligned due to their closely related construction. On the other hand, sentiment variable again shows negligible correlations with the financial variables, with coefficients close to zero and a negative correlation with the return series(-0.0335).

The similarity between the Pearson and Spearman results indicates that the weak relationships observed are not primarily driven by extreme outliers, but instead, they suggest that sentiment does not exhibit a strong contemporaneous monotonic relationship

with Bitcoin volatility or returns, implying that any predictive influence of sentiment is more likely to be through non-linear relationships or lag structures.

4.4 Model estimation results

4.4.1 Model specification and estimation setup

Table 4.9: *Model specifications and estimation setup.*

Setting	Value
Return variable	log_return
Squared returns	Squared returns (r_t^2)
Lagged sentiment variable	Search_Interest_lag
Benchmark model	GARCH Normal
GARCH order (p, q)	(1,1)
GARCH innovations compared	normal, t
RF baseline features	lag_sq_return_1, lag_sq_return_5, lag_abs_return_1, lag_return_1, lag_log_volume_1
RF sentiment-augmented features	lag_sq_return_1, lag_sq_return_5, lag_abs_return_1, lag_return_1, lag_log_volume_1, lag_sentiment_1, lag_sentiment_5
In-sample size	2513
Out-of-sample size	629

Table 4.9 summarises the empirical modeling framework adopted in this study where log returns are used as the primary return variable, while squared returns serve as the proxy for realised volatility given that the study did not use intra-day data, and search interest obtained from Google Trends is employed as the sentiment variable.

Within the econometric modeling framework, GARCH(1,1) and IGARCH(1,1) models are estimated as baseline models under two alternative innovation distributions: the Gaussian and the Student-t. The Student-t specification allows the conditional variance model to accommodate the heavy-tailed behaviour observed in cryptocurrency returns which is not captured by by the Normal GARCH models, and the sentiment effects are incorporated through the GARCH-X and IGARCH-X specifications, where lagged search interest enters

the conditional variance equation as an exogenous regressor.

For the machine learning framework, Random Forest regression models are used to forecast volatility with the baseline specification incorporating lagged volatility and return features, while the sentiment-augmented model adds lagged sentiment variables to the baseline RF model.

The dataset contains 3,142 observations in total from January 1, 2017 to September 8, 2025 and of these, 2,513 observations are used for in-sample estimation, while the remaining 629 observations are reserved for out-of-sample forecast evaluation.

4.4.2 Baseline econometric models

GARCH(1,1)-family models

Table 4.10: *Parameter estimates for baseline GARCH models.*

	GARCH Normal				GARCH Student-t			
	Coef.	Std. Err.	t-stat	p-value	Coef.	Std. Err.	t-stat	p-value
α_1	0.1192	0.0142	8.3673	0.0000	0.0857	0.0139	6.1764	0.0000
β_1	0.8402	0.0167	50.3816	0.0000	0.9133	0.0163	56.0583	0.0000
μ	0.1858	0.0552	3.3693	0.0008	0.1215	0.0411	2.9597	0.0031
ν					3.2412	0.1833	17.6798	0.0000
ω	0.6610	0.0995	6.6420	0.0000	0.1727	0.0801	2.1559	0.0311
AIC	5.2894				5.0655			
BIC	5.2971				5.0751			
Log-Likelihood	-8305.5939				-7952.8942			

Under Gaussian innovations, the baseline GARCH(1,1) in table 4.10, the ARCH parameter ($\alpha_1 = 0.1192$) and the GARCH parameter ($\beta_1 = 0.8402$) imply a persistence level of $\alpha_1 + \beta_1 = 0.9594$, indicating highly persistent volatility dynamics with both coefficients being strongly statistically significant with p-values of 0.0000. The conditional mean parameter ($\mu = 0.1858$) as well as the constant term($\omega = 0.6610$) are also statistically significant with p-values of 0.0008 and 0.0000 respectively. The significance of the mean parameter indicates that average returns are non-zero over the sample period, although this effect is typically small in magnitude relative to volatility dynamics, and the 0.0000 p-value for the constant

term which reflects a positive baseline level of conditional variance, ensures that volatility remains strictly positive even in the absence of recent shocks respectively.

When Student-t innovations are assumed as in table 4.10, the ARCH coefficient declines to 0.0857, while the GARCH coefficient increases to 0.9133, yielding a persistence level close to unity ($\alpha_1 + \beta_1 = 0.9990$) with the estimated degrees-of-freedom parameter ($\nu = 3.2411$) significantly confirming heavy-tailed innovations. The intercept term ω decreases substantially under the heavy-tailed specification (0.1727 compared to 0.6612 under Gaussian errors), which means that part of the unconditional variance is absorbed by the fat-tailed distribution.

The model fit improves under Student-t innovations as shown by the increase in the log-likelihood from -8305.5939 under Gaussian errors to -7952.8942 under the Student-t specification, while both AIC and BIC values decline, indicating a better in-sample fit when heavy-tailed innovations are allowed.

IGARCH(1,1)-family models

Table 4.11: *Parameter estimates for baseline IGARCH models.*

	IGARCH Normal				IGARCH Student-t			
	Coef.	Std. Err.	t-stat	p-value	Coef.	Std. Err.	t-stat	p-value
α_1	0.1639	0.0168	9.7414	0.0000	0.0859	0.0132	6.4884	0.0000
β_1	0.8361				0.9141			
μ	0.1802	0.0546	3.2995	0.0010	0.1211	0.0408	2.9662	0.0030
ν					3.2270	0.1337	24.1296	0.0000
ω	0.4935	0.0750	6.5836	0.0000	0.1674	0.0602	2.7836	0.0054
AIC	5.2975				5.0647			
BIC	5.3033				5.0724			
Log-Likelihood	-8319.3774				-7952.6269			

In the IGARCH framework, persistence is constrained to unity by construction and under Gaussian innovations in table 4.11, the ARCH parameter is estimated at $\alpha_1 = 0.1639$ with the corresponding GARCH parameter $\beta_1 = 0.8361$ and the conditional mean parameter ($\mu = 0.1802$) and the variance intercept ($\omega = 0.4935$) are both statistically significant. The unit persistence implied by the IGARCH specification indicates that volatility shocks do not

decay over time, but instead have a permanent effect on the conditional variance and as a result, the model does not exhibit mean reversion in volatility, meaning that periods of high or low volatility can persist indefinitely following a shock.

Under Student-t innovations, the ARCH parameter declines to 0.0859 while the GARCH parameter increases to 0.9141, which is a reflection of a similar trade-off between immediate shock response and long-run persistence as observed in the standard GARCH model. The estimated degrees-of-freedom parameter ($\nu = 3.2270$) again significantly indicates strongly heavy-tailed return innovations.

Consistent with the GARCH results, model fit improves under the Student-t specification. The log-likelihood increases from -8319.3774 under Gaussian errors to -7952.6269 when Student-t innovations are used, while the information criteria decrease accordingly which confirms that allowing for heavy-tailed innovations better captures the distributional characteristics of Bitcoin returns.

4.4.3 Sentiment-augmented econometric models

The sentiment-augmented specifications (GARCH-X(1,1) and IGARCH-X(1,1)) extend the baseline models by including lagged Google Trends search interest as an exogenous regressor in the conditional variance equation as in equation(3.13)

GARCH-X models

Across both innovation assumptions of the sentiment-augmented GARCH models in table 4.12, estimated sentiment coefficient (γ) is effectively zero with p-values of 1.0000, indicating that lagged search interest does not exert a statistically significant effect on conditional variance within the GARCH-X framework.

Importantly, the remaining variance parameters remain very similar to those obtained in the baseline GARCH models given that under the Gaussian innovations, the ARCH and GARCH parameters are $\alpha_1 = 0.1193$ and $\beta_1 = 0.8400$ respectively, implying strong volatility persistence. Under Student-t innovations, the persistence level remains close to unity with $\alpha_1 = 0.1064$ and $\beta_1 = 0.9189$ and the estimated degrees-of-freedom parameter ($\nu = 2.8792$)

Table 4.12: *Parameter estimates for sentiment-augmented GARCH-X models.*

	GARCH-X Normal				GARCH-X Student-t			
	Coef.	Std. Err.	t-stat	p-value	Coef.	Std. Err.	t-stat	p-value
μ	0.1859	0.0553	3.3625	0.0008	0.1136	0.0403	2.8187	0.0048
γ	0.0000	0.0615	0.0000	1.0000	0.0000	0.0377	0.0000	1.0000
ω	0.6621	0.1352	4.8964	0.0000	0.1156	0.0671	1.7222	0.0850
α_1	0.1193	0.0165	7.2361	0.0000	0.1064	0.0225	4.7304	0.0000
β_1	0.8400	0.0212	39.6764	0.0000	0.9189	0.0143	64.2444	0.0000
ν					2.8792	0.1838	15.6662	0.0000
Log-Likelihood	-8302.8688				-7947.4408			
AIC	5.2900				5.0643			
BIC	5.2996				5.0758			

continues to confirm the heavy-tailed nature of Bitcoin returns.

The model fit is also largely unchanged relative to the baseline models as the log-likelihood values remain similar, suggesting that including sentiment does not materially improve in-sample fit within the linear GARCH framework.

IGARCH-X models

Table 4.13: *Parameter estimates for sentiment-augmented IGARCH-X models.*

	IGARCH-X Normal				IGARCH-X Student-t			
	Coef.	Std. Err.	t-stat	p-value	Coef.	Std. Err.	t-stat	p-value
μ	0.1804	0.0546	3.3027	0.0010	0.1204	0.0411	2.9276	0.0034
γ	0.0000	0.0585	0.0000	1.0000	0.0000	0.0348	0.0000	1.0000
ω	0.4940	0.1057	4.6748	0.0000	0.1678	0.0635	2.6405	0.0083
α_1	0.1641	0.0216	7.6136	0.0000	0.0860	0.0139	6.1946	0.0000
β_1	0.8359				0.9140			
ν					3.2268	0.1355	23.8142	0.0000
Log-Likelihood	-8316.6915				-7950.2027			
AIC	5.2981				5.0654			
BIC	5.3058				5.0750			

Consistent with the GARCH-X results in table 4.13, the sentiment coefficient remains effectively zero across both innovation assumptions, with p-values equal to 1.0000 which

implies that as in the case of the GARCH-X variations, lagged search interest does not contribute to explaining conditional variance within the IGARCH framework. Again, the remaining variance parameters are nearly identical to those estimated in the baseline IGARCH models with the ARCH parameter is $\alpha_1 = 0.1641$ with the corresponding GARCH parameter $\beta_1 = 0.8359$ under Gaussian innovations. The ARCH coefficient declines to 0.0860 while the GARCH coefficient increases to 0.9140 under the Student-t innovations, which also indicates highly persistent volatility dynamics.

These results indicate that the specific sentiment proxy used in this study, lagged Google Trends search interest, does not provide incremental explanatory power for conditional variance once past volatility dynamics are accounted for. This, however, does not imply that sentiment is irrelevant for cryptocurrency markets but rather, it suggests that the predictive content of behavioural variables depends critically on how sentiment is measured. In particular, Google Trends captures aggregate search intensity, which is more closely related to investor attention than to directional sentiment and as such, it may combine heterogeneous signals, including curiosity, speculation, and information demand, which do not necessarily translate into a consistent effect on volatility.

4.4.4 Econometric models diagnostics

Residual diagnostics

Table 4.14: *Ljung–Box diagnostics for baseline GARCH and IGARCH models.*

Model	Lag	LB Stat (Resid)	p-value (Resid)	LB Stat (Sq. Resid)	p-value (Sq. Resid)
GARCH-Normal	10	16.3478	0.0120	6.1427	0.4074
GARCH-Normal	20	22.7529	0.1205	8.8038	0.9213
GARCH-Student-t	10	17.4818	0.0037	5.4926	0.3588
GARCH-Student-t	20	24.6652	0.0546	10.2562	0.8033
IGARCH-Normal	10	17.4010	0.0079	6.9735	0.3233
IGARCH-Normal	20	24.3290	0.0826	11.1743	0.7986
IGARCH-Student-t	10	17.5215	0.0036	5.4856	0.3595
IGARCH-Student-t	20	24.7240	0.0538	10.2721	0.8023

Across all models, the Ljung–Box tests in table 4.14 applied to the squared standardized

residuals are insignificant at both lag 10 and lag 20, with p-values ranging from approximately 0.32 to 0.92 which indicates that the conditional variance equations adequately capture the ARCH effects present in the data, leaving no significant autocorrelation in the second moment for all the models.

For the standardized residuals, however, some evidence of short-run autocorrelation is present at lag 10, with p-values well below 0.05 for all of the specifications. However, these effects largely disappear at lag 20, where the p-values increase above conventional significance thresholds of 5% which suggests that any remaining dependence in the conditional mean is weak and limited to very short horizons, thus, showing that the baseline GARCH-family models successfully absorb the volatility clustering observed in the raw returns, with no remaining serial dependence in squared residuals.

Table 4.15: *Ljung–Box diagnostics for sentiment-augmented GARCH-X and IGARCH-X models.*

Model	Lag	LB Stat (Resid)	p-value (Resid)	LB Stat (Sq. Resid)	p-value (Sq. Resid)
GARCH-X Normal	10	16.4055	0.0058	6.1202	0.2947
GARCH-X Normal	20	22.7662	0.0892	8.7819	0.8886
GARCH-X Student-t	10	18.4924	0.0010	5.2821	0.2596
GARCH-X Student-t	20	25.9265	0.0264	10.4887	0.7256
IGARCH-X Normal	10	17.4473	0.0037	6.9548	0.2240
IGARCH-X Normal	20	24.3342	0.0596	11.1578	0.7413
IGARCH-X Student-t	10	17.5971	0.0015	5.4622	0.2431
IGARCH-X Student-t	20	24.7619	0.0370	10.2539	0.7434

The results of the Ljung-Box diagnostics for the sentiment-augmented GARCH-X and IGARCH-X in table 4.15 mirror those obtained for the baseline models since in all specifications, the test results on the squared standardized residuals remain statistically insignificant at both lag 10 and lag 20, with p-values generally above 0.22 and reaching as high as 0.89 which means that the inclusion of sentiment does not introduce variance misspecification.

As with the baseline models, some evidence of short-run autocorrelation appears in the standardized residuals at lag 10 for all the models and stronger, particularly for the Student-t specifications. However, these effects weaken substantially at lag 20, where the p-values exceed the 5% confidence threshold for all the Gaussian innovations and approach this

threshold for the Student-t at 0.0264 for GARCH-X and 0.0370 for IGARCH-X.

ARCH-LM tests

Table 4.16: *ARCH-LM tests for remaining conditional heteroskedasticity in GARCH-family residuals.*

Model	Lag Order	ARCH LM Statistic	p-value
GARCH-Normal	10	6.1852	0.7995
GARCH-Student-t	10	5.4781	0.8570
IGARCH-Normal	10	7.0830	0.7176
IGARCH-Student-t	10	5.4686	0.8578
GARCH-X Normal	10	6.1791	0.8000
GARCH-X Student-t	10	5.2393	0.8746
IGARCH-X Normal	10	7.0709	0.7187
IGARCH-X Student-t	10	5.4622	0.8582

Across all specifications in table 4.16, the null hypothesis of no remaining ARCH effects cannot be rejected since the reported p-values range from approximately 0.72 to 0.87, which implies that no significant conditional heteroskedasticity remains in the residuals after model estimation, hence confirming that the conditional variance equations successfully capture the volatility clustering present in the raw return series. Importantly, the diagnostic outcome is consistent across both the baseline and sentiment-augmented models, as well as across Gaussian and Student-t innovation assumptions at lag 10 which further confirms the adequacy of these econometric specifications.

4.4.5 Random forest model estimation

To provide a flexible nonlinear benchmark for volatility forecasting, Random Forest regression models are estimated using lagged volatility, return, and market activity variables as predictors with the dependent variable as the realised volatility proxied by squared returns, ensuring consistency with the volatility measure used in the econometric models and similar to the econometric models, two specifications, that is, baseline and sentiment-augmented Random Forests are considered. The baseline model includes lagged volatility dynamics, lagged returns, and trading activity variables while the sentiment-augmented specification

extends this feature set by incorporating lagged Google Trends search interest.

The model estimation is conducted using time-series cross-validation to preserve the temporal ordering of observations, implementing a five-fold rolling cross-validation scheme using the `TimeSeriesSplit` procedure and the hyperparameters are selected through grid search over the number of trees, maximum tree depth, and minimum leaf size. The candidate grid includes 500 and 1,000 trees, maximum depths of 6, 8, and 10, and minimum leaf sizes of 3 and 5 observations and the model selection is based on minimising mean squared prediction error within the cross-validation folds.

To ensure comparability with the econometric models, the Random Forest estimation sample is aligned with the same in-sample and out-of-sample split used in the GARCH-like models. Thus, the training sample contains 2,513 observations, while 629 observations are reserved for out-of-sample forecast evaluation.

In-sample fit

Table 4.17: *In-sample fit statistics for the Random Forest volatility models.*

Model	MSE	RMSE	MAE	R^2
RF Baseline	2498.5390	49.9854	17.7327	0.1801
RF + Sentiment	2492.9679	49.9296	17.7860	0.1819

The baseline RF specification as in table 4.17 produces a mean squared error (MSE) of 2498.5390 and a root mean squared error (RMSE) of 49.9854, with a mean absolute error (MAE) of 17.7327 and an R^2 of 0.1801 which indicates that the model captures only a modest share of the variation in squared returns, suggesting that a substantial portion of volatility dynamics remains unexplained within this specification.

Including sentiment variables leads to only marginal changes in the in-sample fit with the sentiment-augmented model just slightly reducing the MSE to 2492.9676 and the RMSE to 49.9296, while the MAE increases marginally to 17.7860 and similarly, the coefficient of determination increases only slightly by 0.0018 to $R^2 = 0.1819$. Therefore, the overall differences between the two specifications are economically small, implying that sentiment variables contribute only limited additional explanatory power in the in-sample Random

Forest estimation.

The weak in-sample fit of the Random Forest models suggests that the nonlinear tree-based specification is not extracting a strong additional signal from the selected predictors beyond what is already contained in recent volatility dynamics and the negligible difference between the baseline and sentiment-augmented specifications further indicates that lagged Google Trends search interest contributes very little incremental information within this framework.

Feature importance

Table 4.18: *Feature importance rankings for the Random Forest volatility models.*

Model	Feature	Importance
RF + Sentiment	lag_abs_return_1	0.1958
RF + Sentiment	lag_sq_return_1	0.1927
RF + Sentiment	lag_sq_return_5	0.1878
RF + Sentiment	lag_log_volume_1	0.1278
RF + Sentiment	lag_return_1	0.1191
RF + Sentiment	lag_sentiment_5	0.1144
RF + Sentiment	lag_sentiment_1	0.0624
RF Baseline	lag_sq_return_5	0.2538
RF Baseline	lag_abs_return_1	0.2220
RF Baseline	lag_sq_return_1	0.2082
RF Baseline	lag_log_volume_1	0.1754
RF Baseline	lag_return_1	0.1406

In the sentiment-augmented specification, lagged volatility measures dominate the importance scores as shown in table 4.18. The most influential predictors are the lagged absolute return (importance = 0.1958), the first lag of squared returns (0.1927), and the five-day lag of squared returns (0.1878). On the other hand, the sentiment variables contribute comparatively less to the predictive structure of the model, with the five-day sentiment lag ranking sixth (importance = 0.1144) and the one-day sentiment lag ranking seventh (importance = 0.0624), values that are noticeably lower than the importance scores associated with volatility and return dynamics.

In the baseline model, the lagged volatility variables again dominate the ranking, with the five-day squared return lag (0.2538) ranking first, followed by the lagged absolute return

(0.2220), and the one-day squared return lag (0.2082) in third place, providing the largest contributions to predictive performance.

4.5 Out-of-sample forecasting performance

4.5.1 Forecast loss comparison

Table 4.19: *Out-of-sample volatility forecast performance under QLIKE loss, with mean loss differentials relative to the GARCH-Normal benchmark.*

	QLIKE	Rank	Mean Loss Differential
GARCH-Normal	1.7459	1	0.0000
IGARCH-Normal	1.7971	2	-0.0512
GARCH-Student-t	1.8024	3	-0.0565
IGARCH-Student-t	1.8102	4	-0.0643
IGARCH-X Normal	1.8534	5	-0.1075
GARCH-X Normal	1.8733	6	-0.1275
IGARCH-X Student-t	1.9307	7	-0.1848
RF Baseline	1.9847	8	-0.2388
RF + Sentiment	1.9899	9	-0.2440
GARCH-X Student-t	1.9915	10	-0.2457

The out-of-sample forecasting performance of all the models is ranked using the QLIKE loss function as reported in table 4.19, where lower values indicate better predictive accuracy. The baseline GARCH-Normal model achieves the lowest QLIKE loss of 1.7459 and therefore serves as the benchmark specification, followed by IGARCH-Normal (1.7971) and GARCH-Student-t (1.8024). Among the sentiment-augmented specifications, IGARCH-X Normal ranks first (1.8534), though it comes fifth in the overall ranking of the models and GARCH-X Normal sixth (1.8733). Notably, the GARCH-X Student-t model records the

highest QLIKE value among all the model specifications in the study(1.9915, rank 10), and is thus the worst performing model.

In contrast, the Random Forest models produce substantially higher forecast losses, with the baseline Random Forest recording a QLIKE of 1.9847 (rank 8) and the sentiment-augmented version 1.9899 (rank 9). As shown by the mean loss differentials in the final column, both RF models exhibit loss increases of approximately 0.24 relative to the GARCH-Normal benchmark. This means that the nonlinear machine-learning approach does not translate into improved forecast accuracy for daily Bitcoin volatility within the evaluation sample.

Across the out-of-sample comparison, the results favour the econometric models over the Random Forest alternatives as the strongest performance is achieved by the standard GARCH specification under Gaussian innovations, while the RF models rank consistently below the GARCH-family forecasts. This pattern indicates that, for one-step-ahead Bitcoin volatility forecasting with squared returns as the evaluation proxy, explicitly modeling conditional heteroskedasticity and volatility persistence is more effective than relying on a flexible nonparametric regression alone.

4.5.2 Forecast dynamics

GARCH-family volatility forecasts

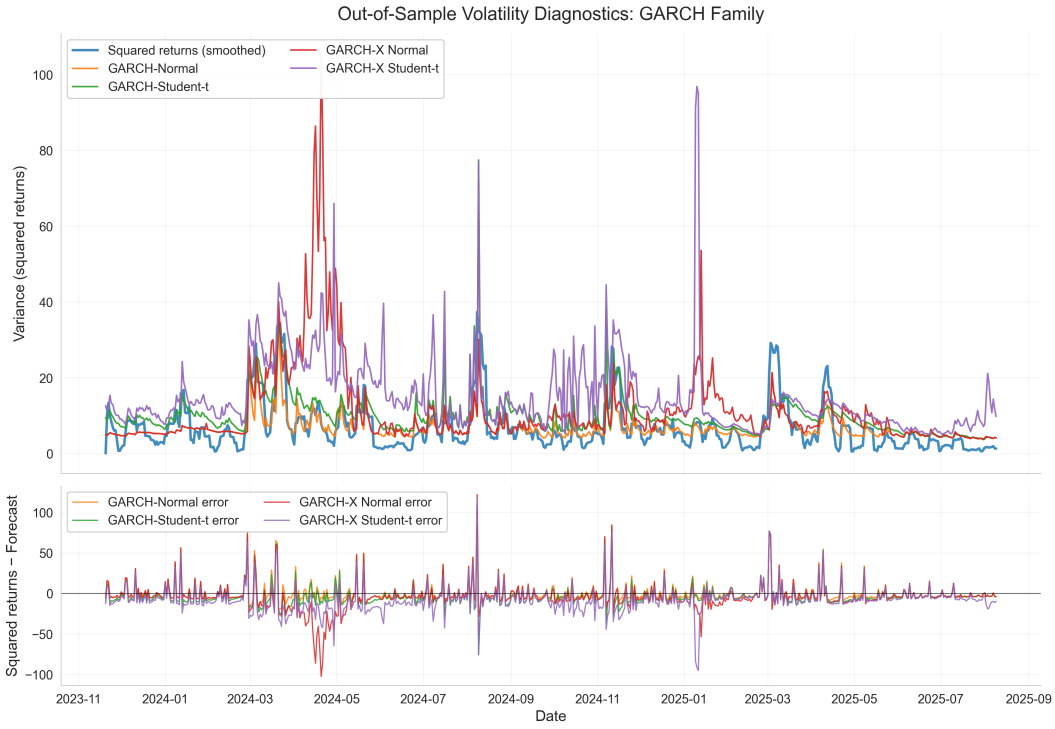


Figure 4.5: *GARCH and GARCH-X forecasts*

Figure 4.5 presents the out-of-sample volatility diagnostics for the GARCH-family models in which the upper panel displays the volatility forecasts generated by the different GARCH specifications together with the squared returns based on squared returns, while the lower panel shows the corresponding forecast errors defined as the difference between realised and predicted volatility.

The squared returns series exhibits pronounced spikes and short-lived bursts of volatility, which affirms the clustering behaviour in cryptocurrency markets, and specifically, in Bitcoin. In this plot, the GARCH-family forecasts track the general level and persistence of volatility reasonably well with occasional higher spikes as the models respond to periods of elevated volatility by gradually increasing the conditional variance forecasts and subsequently declining as market conditions stabilise, a feature of the autoregressive structure embedded in the conditional variance equation.

The forecasts, however, appear noticeably smoother than the squared returns series and tend to underreact to extreme spikes with the plot showing that several large volatility bursts, particularly around mid-2024 and early-2025, are only partially captured by the model forecasts.

The forecast errors in the lower panel tend to spike during periods of abrupt volatility increases, which indicates that the models underestimate the magnitude of extreme volatility shocks. Outside these periods, however, the errors remain relatively small and centred around zero, which shows that the models capture the general volatility level reasonably well, with relatively small differences between model specifications, though the Student-t based models occasionally generate slightly higher volatility forecasts during turbulent periods which reflects their heavier-tailed distributional assumption. Furthermore, the sentiment-augmented GARCH-X specifications display only minor deviations from the baseline GARCH forecasts, a confirmation for the earlier results in this study, indicating that the sentiment variable provides only limited additional explanatory power.

IGARCH-family volatility forecasts

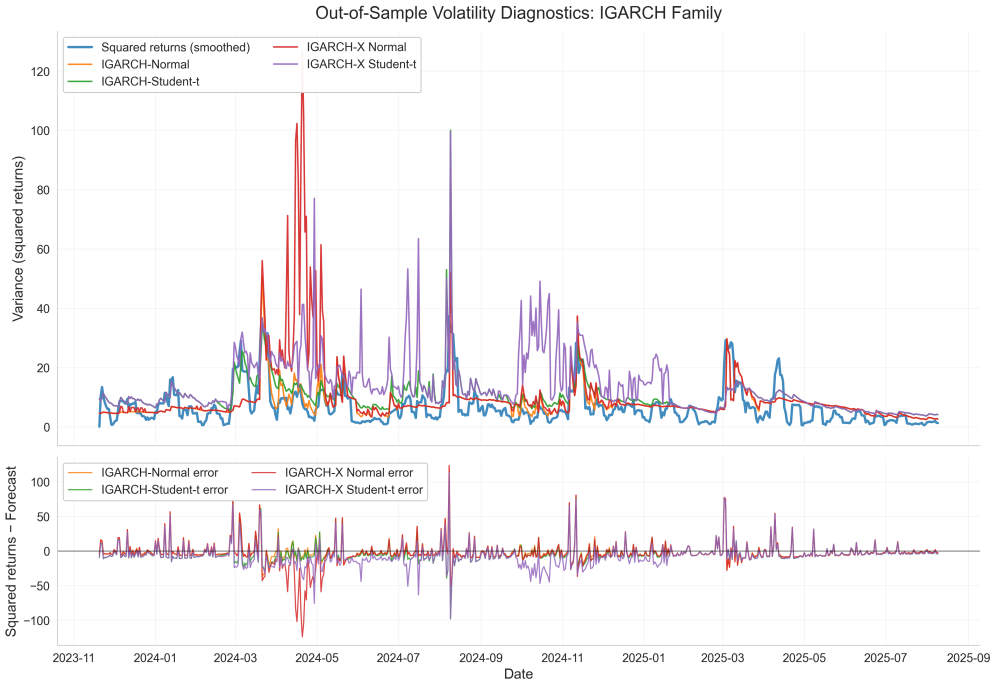


Figure 4.6: *IGARCH and IGARCH-X forecasts*

Similar to figure 4.5, figure 4.6 shows the volatility forecasts generated by the IGARCH-family model specifications in the upper panel and the corresponding forecast errors in the lower panel. The volatility series, similarly, exhibit pronounced spikes and clear clustering behaviour, reflecting the episodic nature of volatility in cryptocurrency markets. Due to the unit persistence constraint imposed in the IGARCH specification, however, volatility shocks tend to decay more slowly compared with the standard GARCH models and consequently, following major volatility spikes, the IGARCH forecasts remain elevated for longer periods even after the squared returns begins to decline, producing smoother forecast dynamics and slower adjustments to changing market conditions. The forecast errors in the lower panel exhibit behaviour similar to the GARCH-family models in figure 4.5.

Random Forest volatility forecasts

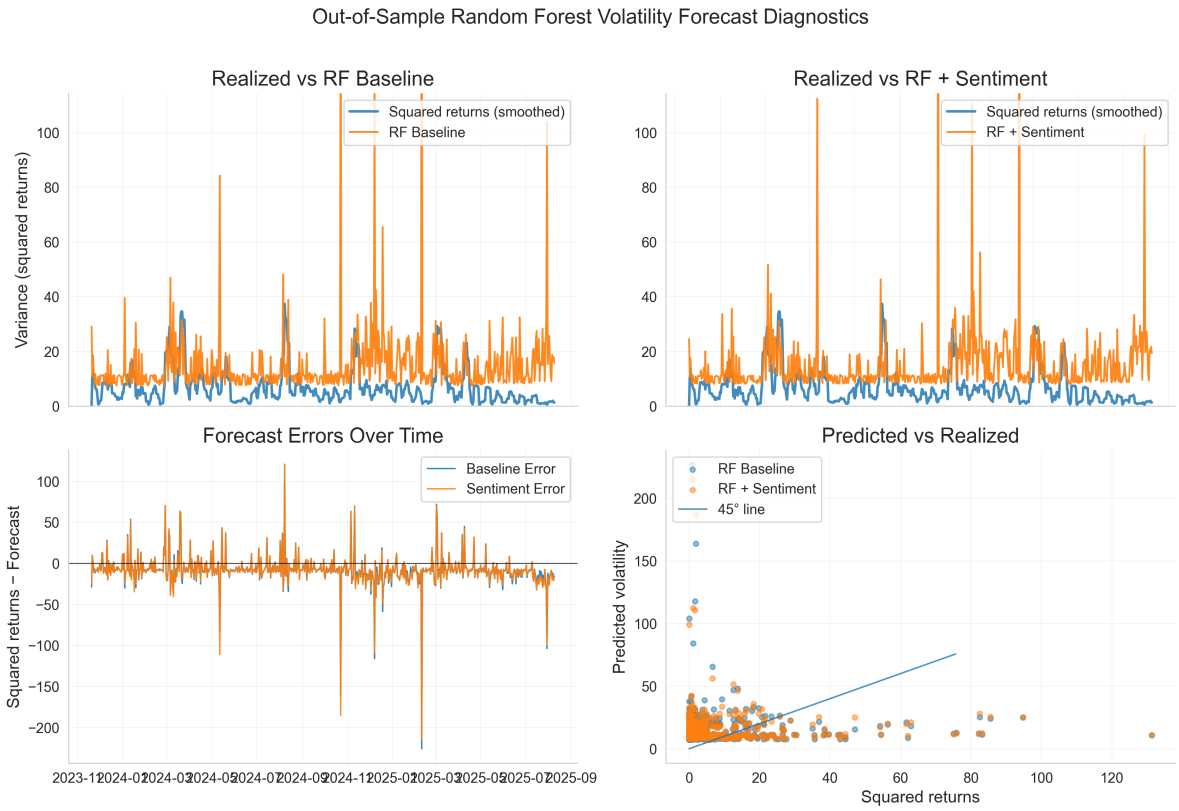


Figure 4.7: *RF models forecast diagnostics*

Figure 4.7 presents diagnostic plots for the out-of-sample volatility forecasts generated by the Random Forest models in which panels (a) and (b) compare the squared returns with the forecasts produced by the baseline Random Forest model and the sentiment-augmented Random Forest model, respectively.

Both models, despite tracking correctly the general direction of volatility movements exhibit substantially higher variability than the squared returns series with both models generating sharper spikes compared to the GARCH-like models that exceed the magnitude of the observed volatility, particularly during periods of market turbulence.

In panel (c), the forecast errors overtime are shown, and even though most errors fluctuate around zero, the distribution is characterised by occasional large positive and negative deviations, which aligns with the substantial overprediction or underprediction of volatility particularly during extreme market episodes.

Panel (d) plots predicted volatility against squared returns where dispersion of points away from the 45-degree line indicates considerable forecast noise, with many observations clustered at relatively low squared returns levels but associated with higher predicted volatility which suggests that the Random Forest models tend to overreact to certain predictor signals and generate exaggerated forecasts even during otherwise moderate volatility conditions. Here, the baseline and the sentiment-augmented models produce very similar forecast paths throughout the evaluation period and this visual similarity confirms the results of the feature importance analysis which showed that sentiment variables contribute relatively little predictive information relative to lagged volatility measures.

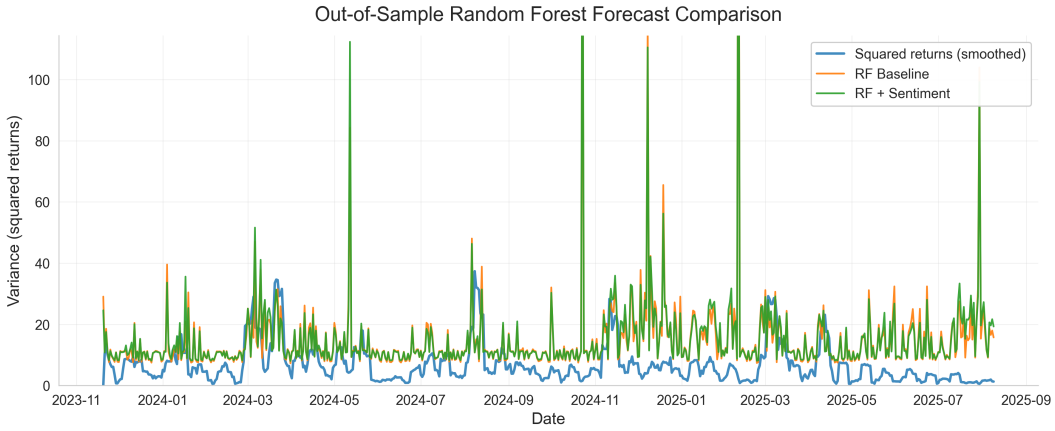


Figure 4.8: *Random baseline vs sentiment models*

In figure 4.8, a visual comparison of the Random Forest models reveals that both models generate volatility forecasts that move closely together over the entire out-of-sample period, with the forecasts exhibiting near-identical responses to the volatility spikes. In particular, most large forecast spikes occur simultaneously in both models, suggesting that they are primarily driven by the underlying return dynamics rather than by additional information contained in the sentiment measure.

4.6 Robustness checks

4.6.1 Superior predictive analysis(SPA) test

Table 4.20: *Superior Predictive Ability (SPA) test relative to the GARCH-Normal benchmark.*

Statistic	p-value	Observations	Models Tested	Benchmark	Block Length	Bootstrap Replications
-1.2849	0.6590	629	9	GARCH-Normal	10	1000

The results of the Superior Predictive Ability (SPA) test are reported in table 4.20 evaluates whether any of the competing forecasting models significantly outperforms the chosen benchmark model, which in this case is the GARCH-Normal specification. The reported SPA statistic is negative (-1.2849) with an associated bootstrap p-value of 0.6590 based on 1000 bootstrap replications and a block length of 10. Since the p-value is well above conventional significance levels, the study fails to reject the null hypothesis that the benchmark model is not inferior to the competing models, meaning that none of the alternative forecasting models provides a statistically superior predictive performance in comparison to the GARCH-Normal benchmark when evaluated using the chosen loss function. Although some models may exhibit marginal improvements in point forecast accuracy, these differences are not large enough to be considered statistically significant within the SPA testing framework.

Illustrated in figure 4.9 is the bootstrap distribution of the SPA statistic obtained from the resampling procedure in which the histogram shows the empirical distribution generated from the bootstrap replications, while the vertical dashed line represents the observed SPA

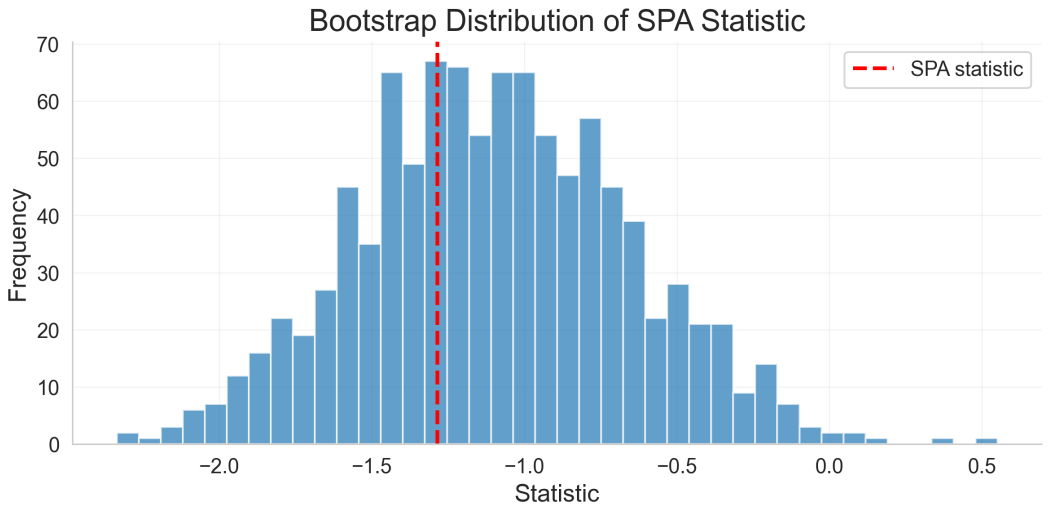


Figure 4.9: *SPA statistic bootstrap distribution*

statistic computed from the original sample. The observed statistic lies well within the central region of the bootstrap distribution rather than in the extreme tail which provides a visual evidence that is consistent with the relatively large p-value reported in table 4.20 and further supports the conclusion that there is no statistically significant evidence that any of the competing models outperform the benchmark model.

4.6.2 Model confidence set(MSC)

Table 4.21: *Model Confidence Set (MCS) inclusion results based on the out-of-sample QLIKE loss matrix.*

Model	Included in Final MCS
GARCH-Normal	Yes
GARCH-Student-t	Yes
GARCH-X Normal	Yes
GARCH-X Student-t	Yes
IGARCH-Normal	Yes
IGARCH-Student-t	Yes
IGARCH-X Normal	Yes
IGARCH-X Student-t	Yes
RF Baseline	Yes
RF + Sentiment	Yes

Table 4.21 reports the inclusion results from the Model Confidence Set (MCS) procedure,

based on the out-of-sample loss matrix, which evaluates whether any forecasting model can be statistically eliminated from the set of superior models according to the selected loss function.

The results indicate that all ten models remain included in the final Model Confidence Set, meaning that none of the competing models can be statistically rejected as inferior relative to the others at the chosen confidence level. In other words, although individual models may exhibit differences in forecast accuracy, these differences are not sufficiently large enough to justify excluding any model from the set of potentially best-performing specifications which suggests that both the traditional econometric models and the Random Forest models forecasting performance across model classes is broadly comparable within the evaluated sample period.

To further confirm this, the summary of the final MCS at the 90% confidence level in table 4.22 retains all the ten forecasting models which means that the predictive performance differences among the competing models are statistically indistinguishable within the sample, which also justifies the SPA results in table 4.20 which similarly failed to identify any model with significantly superior predictive ability relative to the benchmark.

Table 4.22: *Model Confidence Set summary.*

Confidence Level	Models Retained	Final MCS
90%	10	GARCH-Normal, GARCH-Student-t, GARCH-X Normal, GARCH-X Student-t, IGARCH-Normal, IGARCH-Student-t, IGARCH-X Normal, IGARCH-X Student-t, RF Baseline, RF + Sentiment

4.6.3 Alternative volatility proxy

Absolute returns

The forecast evaluation results in table 4.23 are obtained using absolute returns as an alternative volatility proxy. Since the GARCH and IGARCH models generate forecasts for conditional variance, these forecasts are converted to the volatility scale by taking square roots to ensure comparability with the absolute-return proxy.

Table 4.23: *Forecast evaluation under the alternative proxy.*

Model	MSE	RMSE	MAE	Rank	Mean Loss Differential
GARCH-Normal	3.7396	1.9338	1.6149	1	0.0000
IGARCH-Normal	4.2515	2.0619	1.7119	2	-0.5119
GARCH-Student-t	4.8454	2.2012	1.8804	3	-1.1058
IGARCH-Student-t	4.9916	2.2342	1.9139	4	-1.2520
IGARCH-X Normal	5.6646	2.3800	1.8966	5	-1.9250
GARCH-X Normal	5.8440	2.4174	1.9803	6	-2.1044
IGARCH-X Student-t	6.9480	2.6359	2.2456	7	-3.2085
GARCH-X Student-t	7.5688	2.7511	2.3789	8	-3.8292
RF Baseline	7.6986	2.7746	2.3545	9	-3.9590
RF + Sentiment	7.8334	2.7988	2.3776	10	-4.0938

The results show that the overall ranking of the models remains broadly consistent with the baseline QLIKE evaluation. The GARCH-Normal model achieves the best forecasting performance across MSE, RMSE, and MAE, followed by IGARCH-Normal and GARCH-Student-t. The Random Forest models continue to exhibit substantially weaker performance under this alternative proxy, ranking ninth (RF Baseline) and tenth (RF + Sentiment) respectively. This robustness check further confirms that the machine-learning models are less effective than the parametric GARCH-family specifications, even when an alternative volatility proxy is employed.

Alternative evaluation window (Restricted OOS)

The forecast evaluation results under the restricted out-of-sample evaluation window, as reported in table 4.24, show that the model rankings remain broadly consistent with the primary QLIKE-based evaluation even after excluding the earlier portion of the forecasting period. The GARCH-Normal model continues to achieve the best overall performance under both RMSE and MAE. In contrast, the Random Forest models consistently occupy the bottom positions (ranks 9 and 10), recording substantially larger forecast errors compared to the parametric GARCH-family specifications.

The inclusion of the Google Trends sentiment variable does not materially improve forecasting accuracy under this restricted window, which further confirms the stability of

Table 4.24: *Forecast performance under a restricted out-of-sample evaluation window.*

Model	RMSE	MAE	RMSE Rank	MAE Rank	RMSE Loss Differential
GARCH-Normal	12.7346	7.0477	1	1	0.0000
IGARCH-Normal	13.4914	7.8866	2	2	-0.7568
IGARCH-X Normal	13.5705	8.0320	3	3	-0.8359
GARCH-X Normal	13.7141	8.5383	4	4	-0.9796
GARCH-Student-t	13.8265	8.6180	5	5	-1.0919
IGARCH-Student-t	14.2929	8.9795	6	6	-1.5583
IGARCH-X Student-t	17.3557	11.8707	7	7	-4.6211
GARCH-X Student-t	17.7461	12.1941	8	8	-5.0115
RF Baseline	22.2508	13.6849	9	9	-9.5162
RF + Sentiment	22.2978	13.8799	10	10	-9.5632

the main findings across different evaluation periods.

Robustness of sentiment effects

To examine whether the predictive contribution of investor attention might be sensitive to the precise timing of the information, the forecast performance is re-evaluated when the sentiment variable is lagged by an additional day (i.e., using S_{t-2} instead of S_{t-1}). As reported in table 4.25, the results remain highly consistent with the baseline evaluation. The GARCH-Normal model continues to achieve the lowest RMSE and MAE, followed by IGARCH-Normal and GARCH-Student-t. The Random Forest models again occupy the bottom two positions (ranks 9 and 10), recording substantially larger forecast errors than any of the parametric specifications. Importantly, altering the lag structure of the Google Trends variable does not improve the performance of any sentiment-augmented model. Both the GARCH-X and IGARCH-X specifications with the extra lag continue to underperform their non-sentiment counterparts, and the sentiment-augmented Random Forest model remains the worst-performing specification overall. This robustness check reinforces a central finding of the thesis that the broad Google Trends search interest proxy for investor attention provides little to no incremental predictive power for next-day Bitcoin volatility, regardless of the exact timing of the sentiment signal. Even when the information set is shifted by one additional day, the dominant role of lagged volatility dynamics persists,

Table 4.25: *Forecast performance when the evaluation sample is restricted to periods where sentiment lagged by an additional day is available.*

Model	RMSE	MAE	RMSE Rank	MAE Rank	RMSE Loss Differential
GARCH-Normal	13.4567	7.8006	1	1	0.0000
IGARCH-Normal	14.1781	8.4996	2	2	-0.7214
GARCH-Student-t	14.3920	9.4770	3	3	-0.9354
IGARCH-Student-t	14.7127	9.7388	4	4	-1.2561
GARCH-X Normal	17.5614	10.8538	5	6	-4.1047
IGARCH-X Student-t	17.6768	12.5577	6	7	-4.2202
GARCH-X Student-t	18.3841	13.4266	7	9	-4.9274
IGARCH-X Normal	18.5090	10.4878	8	5	-5.0523
RF Baseline	20.9163	13.3543	9	8	-7.4596
RF + Sentiment	21.1539	13.5508	10	10	-7.6972

while the contribution of the sentiment variable stays negligible. These results suggest that more granular or directionally nuanced sentiment measures may be needed if behavioural variables are to meaningfully enhance volatility forecasts in this setting.

Chapter 5: Discussion

5.1 Interpretation of findings

5.1.1 Stylised facts of Bitcoin volatility

The empirical results confirm that Bitcoin volatility exhibits the key stylised facts commonly observed in financial time series, including volatility clustering, strong persistence, and pronounced leptokurtosis. In particular, the estimated persistence parameters from the GARCH(1,1) models (table 4.10), with $\alpha_1 \approx 0.12$ and $\beta_1 \approx 0.84$, imply a persistence level close to unity ($\alpha_1 + \beta_1 \approx 0.96$). This high degree of persistence is even more evident in the IGARCH(1,1) specifications (table 4.11) and is consistent with the strong performance of IGARCH models in the QLIKE-based out-of-sample ranking (table 4.19). These findings directly confirm the autoregressive structure of conditional variance documented in the ARCH and GARCH literature.

These features are central to understanding the model comparison results reported in this study. The strong volatility clustering and persistence observed in Bitcoin returns favour econometric models such as GARCH, which are explicitly designed to capture time-varying variance dynamics. By contrast, models that do not impose a volatility structure, such as Random Forests, may struggle, as is the case in this study, to fully account for these persistent conditional variance patterns as reported in table 4.19.

The distributional properties of the residuals point to the presence of fat tails, implying that extreme movements in Bitcoin returns occur more frequently than would be predicted under a normal distribution. This is in line with the finding of Bollerslev et al. (1994) for the wider financial econometric field and has been widely documented in key cryptocurrency studies as in Chu et al. (2017), Katsiampa (2017), and Ardia et al. (2019). The excess kurtosis of 11.6638 for log returns and 1093.4381 for squared returns in table 4.1, together with the strong rejection of normality in the Jarque–Bera tests (table 4.4), provide direct

statistical confirmation of this stylised fact. On the other hand, the superior performance of the Gaussian model specification relative to the Student-t specifications in the QLIKE ranking (table 4.19) questions the importance of accommodating heavy-tailed behaviour when modeling Bitcoin volatility in this particular sample.

The evidence on asymmetry appears to be weaker or less consistent than in traditional equity markets. While volatility responses to negative returns are well documented for equities (Nelson, 1990), the results in this study show that such effects are either absent or less pronounced in Bitcoin markets. This is consistent with the findings of previous studies such as Katsiampa (2017) and Cheikh et al. (2020), who both reported mixed or limited evidence of leverage effects in cryptocurrency returns. The continued outperformance of the symmetric GARCH-Normal model over asymmetric or Student-t alternatives as in tables 4.19 and 4.23 provides further empirical support for this conclusion.

The results also show that Bitcoin volatility may exhibit periods of structural instability and regime-dependent behaviour. Compared to traditional asset classes, Bitcoin is characterised by rapid changes in market participation, evolving regulatory frameworks, and shifts in investor composition—features which can lead to time variation in volatility dynamics, as documented by Dyhrberg (2016) and Ardia et al. (2019). Thus, the relatively high persistence observed in the estimated models (tables 4.10 and 4.11) may partly capture underlying regime shifts rather than purely stationary volatility dynamics. This interpretation is supported by the Model Confidence Set procedure (table 4.21), which retains *all ten models* at the 90% confidence level, and by the SPA test (table 4.20), which fails to identify any model as statistically superior to the GARCH-Normal benchmark. The robustness of the rankings across the full sample and the restricted out-of-sample window (table 4.24) further reinforces the presence of regime-dependent behaviour.

5.1.2 Econometric model results and volatility persistence

One of the most striking features of the econometric model results is the consistently high degree of persistence in the conditional variance across all GARCH specifications. In the estimated GARCH(1,1) models (table 4.10), the sum of the ARCH and GARCH parameters ($\alpha_1 + \beta_1$) is consistently close to unity (approximately 0.96), suggesting that shocks to

volatility tend to dissipate only gradually. This finding is in line with the classic results of Engle & Bollerslev (1986) for traditional financial time series. At the same time, the observed persistence may be somewhat overstated in the sense that it could be absorbing structural changes in the underlying data-generating process rather than reflecting purely stable dynamics.

The IGARCH(1,1) specifications (table 4.11), which by construction impose a unit root in the volatility process, further reinforce the picture of high persistence. While IGARCH models can sometimes provide a better fit in highly persistent environments, the results in this study show that they do not offer a clear advantage over the standard GARCH models. In fact, the best-performing IGARCH specification is IGARCH-Normal, which ranks second in the QLIKE-based out-of-sample evaluation reported in table 4.19 while the IGARCH-X Normal ranks fifth. This suggests that while volatility in Bitcoin is highly persistent, it is perhaps not quite to the extent implied by fully integrated processes.

The comparison between Gaussian and Student-t innovations also yields a clear pattern as all the Student-t specifications rank below their corresponding Gaussian counterparts in the QLIKE ranking (table 4.19). This suggests that, although fat tails are clearly present in the data as reported in table 4.1 and the Jarque–Bera tests in table 4.4, their practical importance for out-of-sample forecasting may depend more on the evaluation metric and sample period than is commonly assumed.

5.1.3 The role of investor sentiment

The inclusion of lagged Google Trends search interest as an exogenous regressor produces statistically insignificant and economically negligible coefficients on the sentiment term, with the coefficient term $\gamma \approx 0$, with p-values equal to 1.000 in all GARCH-X and IGARCH-X specifications in tables 4.12 and 4.13. The same pattern appears in the machine-learning framework where the Random Forest model that includes sentiment features performs only marginally differently and in all cases slightly worse than the baseline Random Forest as in table 4.19 and the feature-importance results.

These findings suggest that the broad search-volume proxy used in this study does not provide meaningful incremental information for next-day volatility forecasting once the

dominant volatility clustering and persistence are already captured by the standard GARCH terms ($\alpha + \beta \approx 0.96$). This is consistent with the extremely high persistence documented throughout the analysis and highlights the limits of aggregate attention measures in highly persistent volatility processes. In the Random Forest models, for example, lagged volatility and return-based variables remain the dominant predictors, while the inclusion of Google Trends leaves model fit almost unchanged.

A possible interpretation is that Google Trends primarily measures investor *attention* rather than sentiment in the strict sense. A surge in search activity can reflect curiosity, confusion, media exposure, panic, or speculative excitement, motives that may have different implications for subsequent volatility. As a result, the same increase in search volume need not correspond to a consistent directional effect in the variance equation. This possibility aligns with the broader literature as Mai et al. (2018) show that not all online signals are equally informative, with the strongest predictive content coming from the "silent majority" rather than visible activity and with Sapkota (2022) who notes that news-based textual sentiment improves volatility forecasts, but this relies on richer content analysis rather than a broad attention proxy as well as with Hoang & Vo (2024) who interpret Google search volume mainly as investor attention and price pressure rather than as a clean measure of sentiment itself.

The present results therefore offer a more balanced reading of the literature. While behavioural variables can matter, their usefulness depends critically on how they are measured. Search-based attention proxies appear too aggregated to capture the qualitative differences across users, motives, and information environments that drive stronger predictive effects in studies based on forums, news text, or richer nonlinear sentiment indices. This helps explain why Google Trends does not materially improve either the econometric models or the Random Forest specifications in this study.

5.1.4 Forecasting performance and model comparison

Bitcoin volatility displays pronounced persistence and clustering, the exact dynamics for which the GARCH models are explicitly designed through the conditional variance equation. By contrast, Random Forests do not impose a volatility structure but must infer

such persistence indirectly from lagged features. This makes them less effective when the underlying process is dominated by stable heteroskedastic dynamics rather than by complex nonlinear interactions alone.

Importantly, the relatively simple standalone Random Forest specification used in this study is not sufficient to outperform well-specified econometric volatility models. This is evident in the QLIKE-based out-of-sample ranking in table 4.19, where both Random Forest variants rank at the bottom, even after modest hyperparameter tuning by increasing the number of estimators and maximum tree depth. The finding is consistent with comparative studies such as Amoanu (2025), which show that ARMA-GARCH achieves the lowest forecast errors while Random Forest and XGBoost, although competitive on some targets, do not fully account for volatility dynamics in the same way as GARCH-type models.

The broader machine-learning literature is more mixed than a simple "machine learning beats GARCH" narrative would suggest. While Basher & Sadorsky (2022) report strong performance for Random Forests in forecasting Bitcoin price direction, their target is directional movement rather than volatility. Similarly, Diane & Brijlal (2024) find that Random Forest outperforms artificial neural networks, but again the comparison is between two machine-learning models rather than against a strong GARCH benchmark. In contrast, the papers that report the largest gains over traditional models typically rely on more sophisticated architectures, especially hybrid GARCH–LSTM designs such as those in Leber & Egyed (2025), rather than standalone tree-based models.

The implication, therefore, is that the weak Random Forest performance in this thesis is not an anomaly but is consistent with the view that volatility forecasting rewards models that encode persistence and conditional variance dynamics explicitly. Where machine learning succeeds most strongly in the literature, it often does so either on different targets or within hybrid systems that retain a GARCH backbone rather than replacing it altogether. Even after modest hyperparameter tuning, the Random Forest models continue to underperform the benchmark GARCH(1,1) specification, and the addition of Google Trends sentiment provides only marginal benefit in the machine-learning framework.

5.1.5 Robustness of results

The robustness checks provide additional confirmation of the main findings. When absolute returns are used in place of squared returns as an alternative proxy for volatility, the relative ranking of the models remains virtually unchanged, with the GARCH-type models continuing to dominate under MSE, RMSE, and MAE as in table 4.23. This consistency is in line with the view that models grounded in the conditional variance structure are particularly well suited to capture the persistence in volatility, as emphasised by Corsi (2009). Consistently, the Random Forest models continue to perform relatively poorly, further suggesting that their flexibility does not compensate for the weaker alignment with the underlying volatility process.

Allowing sentiment to enter the models with an additional lag leads only to marginal changes in model performance and does not improve the relative rankings of the sentiment-augmented models (table 4.25), a finding that is broadly in line with earlier studies suggesting that sentiment effects, while present, can be unstable or highly dependent on the specific modeling choice and measurement of behavioural variables (Sapkota, 2022; Wang et al., 2023).

Under the reduced out-of-sample window starting in May 2024, the relative ordering of the models remains largely the same as in table 4.24 and this stability across different evaluation periods suggests that the superior performance of the GARCH models is not driven by any particular segment of the sample. As noted by Hansen (2005) and extended by the Model Confidence Set procedure of Hansen et al. (2011), robust model comparison should account for sampling variation and model uncertainty. The fact that the rankings do not change materially across sub-samples reinforces the conclusion that the benchmark GARCH(1,1) specification offers stable and reliable forecasting performance for Bitcoin volatility over the 2017–2025 period.

5.2 Comparison with existing literature

The empirical findings of this study are broadly consistent with the cryptocurrency volatility literature, even though much of the earlier work has appeared in specialised or lower-tier

outlets. The strong volatility persistence, limited leverage effects, and heavy-tailed behaviour documented in foundational studies for example Dyhrberg (2016), Chu et al. (2017), and Katsiampa (2017) are clearly reproduced here using a more recent and comprehensive dataset spanning January 2017 to September 2025.

Specifically, the high persistence parameters in both the GARCH(1,1) and IGARCH(1,1) models in tables 4.10 and 4.11, together with the continued strong performance of IGARCH specifications in out-of-sample forecasting in table 4.19, confirm the near-unit-root behaviour that has been a recurring theme since the earliest cryptocurrency volatility studies. Similarly, the limited evidence of asymmetry and leverage effects give that the symmetric GARCH-Normal model consistently ranking first aligns with the findings of Katsiampa (2017) and Cheikh et al. (2020), who reported weak or inverted asymmetry in Bitcoin returns. The pronounced fat tails and extreme kurtosis in the descriptive statistics (table 4.1) and the Jarque–Bera tests (table 4.4) further mirror the distributional features emphasised in the early literature.

Importantly, the insignificance of the Google Trends sentiment coefficient across all GARCH-X and IGARCH-X specifications and the marginal contribution of sentiment in the Random Forest models suggest that broad attention proxies add little incremental predictive power once volatility persistence is properly accounted for. Moreover, the Random Forest specifications underperform the benchmark GARCH(1,1) model even after hyperparameter tuning, a finding that contrasts with some hybrid or deep-learning studies but is consistent with studies that compare standalone machine-learning methods against well-specified econometric benchmarks.

This consistency with earlier core stylised facts, combined with the updated sample that includes major structural events such as the approval of U.S. spot Bitcoin ETFs, strengthens confidence in the robustness of the key empirical regularities of Bitcoin volatility while highlighting the continued usefulness of relatively simple GARCH-type models for forecasting purposes.

5.2.1 Relation to classical volatility modeling literature

The empirical findings of this study can be understood as an extension of the classical volatility modeling framework developed by Engle (1982) (ARCH) and Bollerslev (1986) (GARCH). The dominant patterns observed in Bitcoin volatility, most notably strong persistence and clustering are in clear alignment with the stylised facts that originally motivated these models. In the estimated GARCH(1,1) and IGARCH(1,1) specifications in tables 4.10 and 4.11, the sum of the ARCH and GARCH coefficients $\alpha_1 + \beta_1$ is consistently close to unity, reproducing the high persistence that has been a central feature of the classical literature. In this sense, the results show that even in new and rapidly evolving markets, the basic mechanisms underlying volatility closely resemble those found in more established financial assets.

That said, some aspects of the results are less in line with the traditional equity-market framework. For instance, while negative shocks are typically associated with larger increases in volatility than positive shocks in equity markets (Nelson, 1990), the evidence for such leverage effects in Bitcoin is weaker and less consistent. The symmetric GARCH(1,1)-Normal model consistently outperforms or matches more complex specifications in the out-of-sample QLIKE ranking in table 4.19, confirming the limited relevance of classical asymmetric extensions in this market.

As highlighted by Engle & Bollerslev (1986), allowing for heavy-tailed innovations should improve the statistical fit of volatility models by accommodating excess kurtosis. Although the present results are broadly consistent with this insight as in the Jarque–Bera tests results in table 4.4 and the descriptive statistics in table 4.1, the gains in forecasting performance from using Student-t innovations remain relatively modest. In fact, the standard GARCH(1,1) specification under Gaussian innovations is the strongest performer across the main evaluation and all robustness checks. Given that the classical framework appears not only adequate but also well-equipped to forecast Bitcoin volatility in this sample, the results reinforce the continued relevance of simple GARCH-type models even in the presence of substantial behavioural and structural changes in cryptocurrency markets.

5.2.2 Comparison with cryptocurrency volatility studies

A common result across many studies is that GARCH-type models provide a reasonable description of Bitcoin volatility dynamics as both Dyhrberg (2016) and Chu et al. (2017) documented strong volatility clustering and persistence, with standard GARCH models often performing well relative to alternatives. Subsequent work by Katsiampa (2017) and Ardia et al. (2019) further showed that extensions of the GARCH framework can capture key features of cryptocurrency returns. The results of the present study align well with this literature as the estimated GARCH(1,1) and IGARCH(1,1) models reproduce, while the GARCH models clearly dominate the other specifications considered in the out-of-sample evaluation.

Unlike studies that identify gains from component, smooth-transition, regime-switching, or asymmetric GARCH variants, this thesis finds that a parsimonious Gaussian GARCH(1,1) provides the best out-of-sample volatility forecasts among the models examined. In particular, neither the IGARCH specifications nor Student-t innovations improve forecasting accuracy, despite their stronger persistence and superior in-sample fit. This difference highlights that the relative performance of volatility models depends critically on the forecasting objective, the evaluation metric, and the degree of model complexity employed. The robustness of the GARCH-Normal ranking across the full sample, the restricted out-of-sample window, and the alternative volatility proxy further reinforces this conclusion. Thus, the present analysis confirms the core stylised facts identified in the earlier cryptocurrency volatility literature while demonstrating that, in this longer and more recent sample, a relatively simple GARCH specification remains the most reliable forecasting tool.

5.2.3 Comparison with sentiment and investor attention literature

The results of this study provide only limited support for search-based attention as a useful volatility predictor in the specific form considered here. The coefficient on lagged Google Trends search interest, γ is statistically insignificant and economically negligible in all sentiment-augmented GARCH-X and IGARCH-X specifications in tables 4.12 and 4.13, and the inclusion of sentiment features adds almost no predictive power in the Random Forest models as in table 4.19 and the corresponding feature-importance results. This differs

somewhat from parts of the behavioural finance and cryptocurrency literature.

As Mai et al. (2018) demonstrate, the predictive content of online activity depends heavily on user heterogeneity, highlighting that the signals from the "silent majority" are often informative, whereas activity from highly visible users tends to be less useful. Their findings imply that aggregate attention measures can conceal important differences in information quality. Similarly, Sapkota (2022) finds that news-based textual sentiment improves HAR-RV volatility forecasts, while Hoang & Vo (2024) shows that abnormal Google search volume predicts returns, volatility, and trading volume. More recent work by Leber & Egyed (2025) further supports the idea that predictive gains depend critically on the quality and granularity of the sentiment measure.

Against this background, the failure of Google Trends to improve the present models should not be interpreted as a rejection of behavioural effects altogether. Rather, it provides evidence that this particular proxy is too noisy or too coarse for the one-step-ahead volatility forecasting task once the dominant autoregressive volatility structure is already accounted for. The present results therefore fit a more balanced reading of the literature, highlighting that the behavioural variables can matter, but their usefulness depends critically on how they are measured and on the forecasting objective.

5.2.4 Relation to machine learning forecasting studies

The machine-learning results in this thesis are weaker than those reported in several recent studies, though they are not outside the range of evidence in the literature. A useful starting point is to distinguish between different types of comparisons. Some papers compare Random Forest with simpler statistical or machine-learning benchmarks and find that RF performs well for example in Basher & Sadorsky (2022) who study price-direction forecasting and Diane & Brijlal (2024) in squared returns forecasting. These studies establish that Random Forest can be a strong benchmark in certain contexts, but they do not imply that it should systematically dominate well-specified GARCH-family volatility models.

A smaller but important set of studies shows that GARCH-type models can remain competitive with, or superior to, tree-based ML methods when volatility dynamics are central to the forecasting problem. In particular, Amoanu (2025) finds that

ARMA-GARCH achieves the lowest forecast errors overall, while Random Forest and XGBoost are competitive but do not directly model volatility persistence in the same way. The results of the present study are consistent with this view showing that even after modest hyperparameter tuning, both Random Forest specifications continue to underperform the benchmark GARCH(1,1) model in the QLIKE-based out-of-sample evaluation. Moreover, the addition of Google Trends sentiment provides only marginal benefit in the machine-learning framework.

Machine learning models appear to perform best in their hybrid versions, which combine econometric structure with flexible nonlinear learners. This pattern is clearly visible in studies such as Leber & Egyed (2025), which report superior performance for a sentiment-augmented GARCH-LSTM framework. The weaker standalone Random Forest performance observed here therefore aligns with the broader literature that volatility forecasting rewards models that encode persistence and conditional variance dynamics explicitly, rather than relying solely on data-driven approximation of these patterns.

While a substantial body of cryptocurrency volatility research has appeared in specialised outlets, the present study builds upon and extends this literature by using a longer, more recent sample from 2017 to late 2025, a period that includes major structural events such as the approval of U.S. spot Bitcoin ETFs. This study therefore contributes by providing updated, methodologically rigorous evidence using both classical econometric and machine-learning approaches within a unified forecasting framework.

5.3 Limitations of the study

5.3.1 Data limitations

A fundamental limitation of the present analysis is the use of squared daily returns as a proxy for squared returns. Although this is the conventional approach in the ARCH/GARCH literature (Engle, 1982; Bollerslev, 1986) and in many recent cryptocurrency studies (Sapkota, 2022; Zahid et al., 2022), volatility itself remains inherently latent and unobservable (Patton & Sheppard, 2009). Any proxy therefore introduces measurement

error that can affect both parameter estimation and forecast evaluation. Squared returns, in particular, are known to be quite noisy, especially at daily frequency, and may not fully capture the underlying volatility process. This limitation is partly mitigated by the robustness checks using absolute returns in table 4.23, which yield qualitatively similar rankings, but the core issue of proxy noise remains.

The reliance on daily observations, while necessary for computational feasibility and data availability over a long sample, necessarily obscures important intra-day dynamics. Cryptocurrency markets are known for rapid price movements within a single trading day driven by high-frequency trading, news shocks, and market-microstructure effects. These short-term fluctuations are not captured by open-to-close returns, potentially leading to an incomplete characterisation of volatility behaviour.

Furthermore, Bitcoin markets have undergone substantial structural changes during the 2017-2025 period, including the growing involvement of institutional investors, major regulatory developments, and the approval of U.S. spot Bitcoin ETFs. As noted by Babalos et al. (2025) and Gensler (2024), such events can alter volatility dynamics and introduce new sources of uncertainty. The increasing integration of cryptocurrency markets with the broader financial system (Argente & Van Patten, 2024) may also generate additional volatility spillovers not present in earlier periods. Consequently, the data-generating process is unlikely to have remained stable across the entire sample, and the high persistence observed in the estimated models in tables 4.10 and 4.11 may partly reflect these structural shifts rather than purely stationary dynamics.

Finally, although Bitcoin price and volume data are generally reliable, they are sourced from a decentralised, multi-exchange market with varying levels of liquidity and regulation. This can introduce minor discrepancies in reported prices and volumes, particularly during periods of market stress. While such issues are unlikely to overturn the main conclusions of the study, they may have added noise that affects both the econometric estimations and the performance of the Random Forest models, which are sensitive to data quality and feature construction (Dudek et al., 2024). The focus on a single asset (Bitcoin) also limits the generalisability of the findings, as other cryptocurrencies may exhibit different liquidity, maturity, and investor-composition characteristics.

5.3.2 Sentiment measurement limitations

The present study relies on Google Trends search interest as a proxy for investor attention. While this variable is widely used and publicly available, it captures broad information demand rather than directional sentiment. As a result, the coefficient on lagged search interest, γ is statistically insignificant and economically negligible in all sentiment-augmented GARCH-X and IGARCH-X specifications and the same variable adds almost no predictive power in the Random Forest framework.

This outcome does not imply that behavioural variables are irrelevant in cryptocurrency markets, but rather that the specific proxy employed here may be too coarse to isolate the type of sentiment that matters for volatility formation. Google Trends, being an aggregate measure, cannot distinguish between these different types of participants as silent majority versus vocal minority reported in Mai et al. (2018). A rise in search intensity could reflect optimism, concern, curiosity, media exposure, or panic—motives that may have different implications for subsequent volatility.

Relatedly, search-based measures primarily capture attention rather than behaviour. While attention may precede trading activity, the connection is not guaranteed, as not all individuals who search for information actually participate in the market (Coulter, 2022). Against this background, the weak results obtained with Google Trends should be interpreted as evidence of the limitations of a broad attention proxy rather than a rejection of behavioural effects altogether. More granular sentiment measures derived from Bitcoin-specific forums like BitcoinTalk, social media platforms like Reddit or X(Twitter), or textual analysis of news articles may contain stronger and more stable predictive content for volatility forecasting.

5.3.3 Market structure considerations

Unlike traditional financial markets, which have developed over decades within relatively stable institutional frameworks, cryptocurrency markets remain in a process of rapid transformation. This ongoing evolution makes it more difficult to treat observed relationships as stable over time.

The composition of the Bitcoin market has changed markedly over the sample period. In its early stages, trading was largely dominated by retail investors and speculative traders, contributing to the high volatility documented by Dyhrberg (2016). More recently, increased institutional participation and the introduction of sophisticated financial instruments such as futures, options, and spot ETFs have altered both the level and the dynamics of volatility, as different types of participants respond to information in different ways (Ardia et al., 2019; Babalos et al., 2025). Consequently, patterns observed in earlier periods may not carry over directly to more recent market conditions.

Regulatory developments constitute another important source of structural change. The ongoing discussions around classification, oversight, and the integration of cryptocurrencies into the broader financial system can shape investor expectations and introduce new forms of uncertainty or stability, as noted by Gensler (2024). These regulatory shifts are likely to influence volatility dynamics in ways that are not easily captured by models assuming a stable environment.

Compared with mature equity markets, cryptocurrency markets are still relatively young. Early studies such as Chu et al. (2017) and Katsiampa (2017) highlighted strong volatility clustering and limited stabilising mechanisms. Although liquidity and market infrastructure have improved over time, more recent evidence (Cheikh et al., 2020) suggests that volatility remains elevated and highly sensitive to external shocks. This combination of gradual maturation and persistent instability helps explain why persistence remains high in the estimated models while other features, such as asymmetry, appear less pronounced.

Finally, Bitcoin's growing integration with traditional financial markets like equities, commodities, and foreign exchange introduces additional channels for shock propagation. As Babalos et al. (2025) highlight, these linkages can amplify volatility through spillover effects. Such integration complicates the task of modeling volatility within a single-asset framework, as external shocks from broader markets may exert an influence that is not fully captured by models focused solely on Bitcoin's own past returns and volume.

Chapter 6: Conclusion

6.1 Summary of main findings

6.1.1 Volatility characteristics

The empirical analysis confirms that Bitcoin volatility exhibits the core stylised facts commonly observed in financial time series, including strong volatility clustering, extremely high persistence, and heavy-tailed behaviour. The estimated GARCH(1,1) and IGARCH(1,1) models show persistence parameters ($\alpha_1 + \beta_1$) consistently close to unity, indicating that volatility shocks decay only very slowly. This is further supported by the continued competitive performance of IGARCH specifications in the out-of-sample forecasting.

The periods of elevated volatility tend to persist over time, while extreme movements occur far more frequently than would be predicted under a Gaussian framework. These features are clearly visible in the descriptive statistics and the strong rejection of normality in the Jarque–Bera tests. The pronounced fat tails and the occasionally very large volatility spikes further underscore the importance of models capable of capturing time-varying conditional variance and heavy-tailed innovations in cryptocurrency markets.

6.1.2 Model performance

Across all specifications, the GARCH(1,1) model with Gaussian innovations provides the strongest out-of-sample forecasting performance. It consistently ranks first under the QLIKE loss function and maintains this position across all robustness checks, including the alternative absolute-returns proxy and the restricted out-of-sample window.

Although the IGARCH(1,1) models capture high persistence by construction, they do not offer a clear improvement over the standard GARCH specification and in several cases perform worse. This suggests that imposing full unit-root persistence may be too restrictive

for Bitcoin volatility dynamics in this sample. Similarly, while Student-t innovations improve in-sample fit by accommodating the heavy tails present in the data, these gains do not translate into better forecasting accuracy. The Gaussian specifications consistently outperform their heavy-tailed counterparts in out-of-sample evaluation, indicating that increased distributional flexibility may lead to overfitting rather than improved predictive performance.

The Random Forest models perform substantially worse than the econometric benchmarks. Despite their flexibility in capturing nonlinear relationships, they fail to adequately model the strong volatility persistence that characterises Bitcoin returns. As a result, both Random Forest variants, the baseline and the sentiment-augmented models produce higher forecast errors across all evaluation metrics and rank at the bottom of the QLIKE comparison. Even after modest hyperparameter tuning, the gap between the Random Forest models and the benchmark GARCH(1,1) specification remains significantly large.

6.1.3 Role of sentiment

The inclusion of search-based attention, proxied by lagged Google Trends search interest, does not materially improve forecasting performance in either the econometric or the machine-learning frameworks. In the GARCH-X and IGARCH-X models the coefficient on the sentiment term, γ is statistically insignificant and economically negligible in all specifications. Similarly, the Random Forest model that incorporates sentiment features shows only negligible changes in predictive accuracy relative to the baseline Random Forest and, in some cases, performs marginally worse.

These results suggest that Google Trends data, while a convenient and widely used proxy for investor attention, is too general and aggregated to serve as a reliable predictor of next-day Bitcoin volatility once the dominant autoregressive volatility structure is already accounted for. The study therefore concludes that broad search-volume measures capture attention rather than directional sentiment and add little incremental information beyond what is already contained in lagged returns and volatility.

6.1.4 Robustness of results

The main findings remain stable across a comprehensive set of robustness checks. When absolute returns are used in place of squared returns as an alternative volatility proxy, the relative ranking of the models is virtually unchanged, with the GARCH-type specifications continuing to dominate. Similarly, restricting the evaluation window to the more recent period beginning in May 2024 does not alter the ordering of the models in any material way. And finally, allowing sentiment to enter the models with an additional lag produces only marginal changes in performance and does not improve the relative standing of the sentiment-augmented specifications.

In all three robustness exercises the GARCH(1,1) specification with Gaussian innovations consistently outperforms both the IGARCH variants and the Random Forest models. This stability indicates that the superior forecasting performance of the benchmark GARCH model is not driven by specific modeling choices, particular segments of the sample period, or the choice of volatility proxy. The results are therefore robust and generalisable within the 2017–2025 sample examined in this study.

6.2 Contributions of the study

6.2.1 Empirical contribution

This study provides updated evidence on Bitcoin volatility dynamics using a long and recent sample period spanning January 2017 to September 2025. This period captures major structural developments in the cryptocurrency market, most notably the approval of U.S. spot Bitcoin ETFs. The results demonstrate that a standard GARCH(1,1) model with Gaussian innovations remains the best-performing specification for out-of-sample volatility forecasting, outperforming both IGARCH variants and Random Forest models across the main QLIKE evaluation and all robustness checks. These findings confirm the robustness of core stylised facts of high persistence, volatility clustering, and heavy tails present in cryptocurrency markets, while showing that simple econometric models continue to be highly effective even in a rapidly evolving market.

6.2.2 Methodological contribution

Methodologically, the study contributes by conducting a direct and systematic head-to-head comparison between classical GARCH-family models that is, GARCH, IGARCH, and their sentiment-augmented versions, and Random Forest regressions within a unified forecasting framework. The analysis employs a rolling-window design, evaluates performance using the QLIKE loss function, and applies the Superior Predictive Ability (SPA) test and the Model Confidence Set (MCS) procedure to account for data-snooping and model uncertainty. This ensures a statistically robust and transparent comparison across fundamentally different model classes.

6.2.3 Contribution to sentiment literature

By examining the incremental value of Google Trends search interest, the findings contribute to the growing literature on behavioural finance in cryptocurrency markets. The sentiment coefficient is statistically insignificant and economically negligible in all GARCH-X and IGARCH-X specifications, and the inclusion of sentiment features adds only marginal benefit in the Random Forest framework. This highlights the limitations of broad, aggregate search-volume measures as proxies for investor sentiment and suggests that more specific sentiment indicators may be required to uncover stronger behavioural effects in volatility forecasting.

6.3 Practical implications

6.3.1 Implications for investors and risk management

The results suggest that relatively simple econometric models remain highly effective tools for volatility forecasting in cryptocurrency markets. For practitioners, this implies that a well-specified GARCH(1,1) model can provide reliable inputs for risk management, portfolio allocation, and Value-at-Risk calculations without the need for complex machine-learning frameworks. Given that the GARCH-Normal specification consistently delivers the best

out-of-sample performance across multiple evaluation criteria and robustness checks, investors and risk managers can confidently rely on this parsimonious model for day-to-day volatility forecasting and hedging decisions.

6.3.2 Implications for model selection

The findings also highlight the importance of aligning model choice with the underlying data-generating process. In environments characterised by strong persistence and volatility clustering as is the case in cryptocurrency markets, models that explicitly incorporate conditional variance dynamics such as GARCH tend to outperform more flexible but structure-free alternatives such as standalone Random Forests. This suggests that model structure may be more important than flexibility in volatility forecasting tasks. Practitioners should therefore prioritise models that directly capture the autoregressive nature of volatility rather than relying solely on general-purpose machine-learning algorithms.

6.3.3 Implications for sentiment-based forecasting

The weak performance of Google Trends as a predictor indicates that not all sentiment proxies are equally informative. For practitioners seeking to incorporate behavioural indicators, the results emphasise the need for more refined sentiment measures rather than relying solely on aggregate attention metrics. While investor attention clearly fluctuates over time, the broad search-volume proxy used in this study adds little incremental predictive power once past volatility is already accounted for. Future sentiment-enhanced strategies should therefore focus on higher-quality, context-specific sentiment signals.

6.4 Future research directions

6.4.1 Improved sentiment measures

The present study relies on a broad Google Trends attention proxy that captures general information demand rather than directional investor sentiment. Future research could construct more specific daily sentiment indices from Bitcoin-specific sources such as

BitcoinTalk forum posts, Reddit discussions sites such as r/Bitcoin and r/cryptocurrency, or large news corpora. Such measures may better distinguish between positive, negative, and neutral sentiment and could be compared directly with the aggregate search-volume variable used here to assess whether richer textual sentiment improves volatility forecasts.

6.4.2 Hybrid econometric-machine learning models

The relatively weak performance of standalone Random Forest models in this study suggests that machine learning may be more effective when combined with econometric structure. Future work could therefore explore hybrid models that integrate GARCH-type dynamics or IGARCH for persistence with nonlinear learners such as LSTM, or GRU. These hybrid approaches have shown promising results in recent literature and could provide a better balance between interpretability and predictive accuracy in cryptocurrency volatility forecasting.

6.4.3 High-frequency volatility modeling

The current analysis uses squared daily returns as a volatility proxy, which introduces measurement noise. Future studies could employ high-frequency intraday data to construct more accurate realised volatility measures for example using the Heterogeneous Autoregressive (HAR) framework of Corsi (2009). Comparing these high-frequency proxies against daily squared returns would help determine how much forecast accuracy is lost due to the aggregation of information and whether the relative performance of GARCH versus machine-learning models changes when a cleaner volatility measure is available.

6.4.4 Nonlinear and regime-based models

Given the evidence of structural changes and evolving market conditions over the 2017-2025 period, future research could extend the analysis to explicitly nonlinear econometric models. Promising directions include regime-switching GARCH, smooth-transition GARCH, and component GARCH specifications. These models may better capture time-varying volatility dynamics and potential regime shifts induced by regulatory developments, institutional

inflows, or macroeconomic shocks, thereby providing a more detailed understanding of Bitcoin volatility behaviour.

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Chapter A: Declaration of the Use of AI Tools

This thesis was prepared with limited assistance from generative artificial intelligence (AI) tools.

- CurreChat was used at an early stage to support the development of the overall structure and outline of the thesis.
- ChatGPT was used for limited purposes, including summarising selected academic literature, refining language, and improving clarity and structure during the drafting process.
- Grok was used to review and improve the language, coherence, and logical flow of the document.

All substantive components of the thesis, including the formulation of the research questions, methodological design, data collection, model implementation, empirical analysis, and interpretation of results, were carried out independently by the author. No AI-generated data, models, or empirical results have been used in this study.

As the author, I assume the full responsibility for the accuracy of the content and originality of this thesis work.

Chapter B: Data and Code Availability

All code, data processing scripts, empirical notebooks, LaTeX source files, and generated outputs for this thesis are publicly available in the following GitHub repository:

`https://github.com/eokinyo/Thesis_Work`

The repository is structured as follows:

- `scripts/` — contains the two main working files:
  - The R Markdown script (`Chapter4_GARCH_Family.Rmd`) for estimation and forecasting of all GARCH-family models.
  - The Python Jupyter notebook (`Chapter4_Empirical_Analysis.ipynb`) for data preparation, Random Forest modeling, forecast evaluation, and robustness checks.
- `scripts/WriteUps/` — contains the archived complete LaTeX source files of the thesis, which were written and compiled using Overleaf.
- `Data/` — contains the raw and processed datasets together with the data-fetching scripts:
  - Bitcoin daily price and volume data obtained via the `yfinance` library.
  - Google Trends search interest data collected using the `pytrends` library.
- `outputs/` — contains all tables and figures used in the thesis.

Detailed instructions for reproducing the complete analysis (including package requirements, execution order, and expected outputs) are provided in the `README.md` file located in the root directory of the repository. The repository is designed to ensure full transparency and reproducibility of all empirical results presented in Chapters 4 and 5.